Logistics Risk and Compliance

Course 4 — Logistics Costing, Risk and Cross-Border Operations · Module 11 · Teaching content for the Theory Builder

Review copy. Each lesson lists its blocks in entry order: enter each block's heading and body into the Theory Builder; enter practice questions as Lesson Checks; enter the module assessment via Configure assessment. Grey appendix sections are audit notes, not for entry.

Lesson 11.1 — Cargo Insurance and Risk Allocation

Summary

This lesson introduces cargo insurance and the principles of risk management in logistics operations. Learners will examine common transportation risks, insurance coverage, exclusions, claims procedures, and the allocation of risk between trading parties. The lesson also explains how Incoterms, contractual obligations, and insurance arrangements influence financial responsibility when cargo is lost, damaged, or delayed.

Learning objectives

- Define cargo insurance and explain its importance in logistics.
- Identify common risks affecting goods during transportation and storage.
- Distinguish between cargo insurance and carrier liability.
- Describe common categories of cargo insurance coverage.
- Explain the importance of insurance exclusions and policy conditions.
- Identify how Incoterms allocate transportation risk between buyers and sellers.
- Describe the basic procedures involved in reporting cargo loss or damage.
- Identify documents commonly required when submitting insurance claims.
- Explain how preventive measures and insurance work together to manage cargo risks.

Key terms

Cargo Insurance — Insurance designed to provide financial protection against specified loss of or damage to goods during insured transportation or related activities.

Insured — The person or organisation covered under an insurance policy.

Insurer — The company or organisation providing insurance coverage under an agreed policy.

Insurance Premium — The amount paid for insurance coverage.

Sum Insured — The monetary amount for which an insured interest is covered under the policy.

Insurance Exclusion — A circumstance, cause, or category of loss that is not covered by an insurance policy.

Deductible — The portion of an insured loss borne by the insured under the applicable policy.

All Risks Coverage — Insurance covering fortuitous physical loss or damage except where excluded by the policy terms.

Named Perils Coverage — Insurance covering losses caused by specifically identified risks or events.

Carrier Liability — The legal or contractual responsibility of a carrier for specified cargo loss, damage, or delay.

Risk Transfer — The point or process through which responsibility for specified risks passes from one party to another.

Insurance Claim — A formal request for compensation under an insurance policy.

Survey Report — A document recording the findings of an examination of cargo condition, loss, or damage.

Risk Mitigation — Actions taken to reduce the likelihood or consequences of an identified risk.

Teaching content

KEY CONCEPT

Risk allocation comes before insurance

This lesson opens Module 11, Logistics Risk and Compliance, with the discipline that holds the rest of the module together: managing the possibility that cargo is lost, damaged, or delayed, and knowing in advance who carries the financial consequence. Two different questions are hiding inside that sentence. The first is a question of allocation: if something goes wrong on the journey, which trading party bears the loss? The second is a question of compensation: having borne it, can that party recover the money? Insurance answers only the second question. Insurance compensates; it never assigns responsibility. Responsibility is allocated by the contract of sale, above all by the Incoterms 2020 rule the parties have chosen, which Lesson 4.3, Incoterms and Allocation of Responsibilities, taught in full.

Risk transfer is the point or process through which responsibility for specified risks passes from one party to another. Risk mitigation is the set of actions taken to reduce the likelihood or consequences of an identified risk. With those two ideas in place, the professional sequence for managing cargo risk runs in three steps.

1. Identify the risk. Ask what can happen to these particular goods on this particular route: physical loss, damage, the consequences of delay, even a share in another party's maritime sacrifice.

2. Allocate the risk. Ask which party carries each risk, and from which precise point. The trade term answers this before anything goes wrong.

3. Insure the risk. Ask whether the party carrying the risk can afford to carry it unprotected, and buy cover accordingly.

Prevention and insurance are partners, not rivals: mitigation lowers the chance of a claim, and insurance stands behind the losses that mitigation could not prevent. The blocks that follow walk the sequence in order.

KEY CONCEPT

What can happen to goods on the move

Before anyone buys cover, the risk must be identified, and the risks are not the same for every consignment. The common families are these.

1. Physical loss. Goods may disappear entirely: a container lost overboard, a whole consignment never delivered, part of a delivery missing, or goods stolen in transit or from storage.

2. Physical damage. Goods may arrive but arrive ruined or impaired: crushed by bad handling, soaked by water entering a container, burnt in a fire, damaged when a vehicle collides or overturns, or spoiled by the weather on an exposed quay.

3. Delay and its consequences. A consignment may arrive late with consequences that are financial rather than physical: a perishable cargo past its market, a factory line stopped for want of parts, a seasonal order missed. Delay deserves separate treatment because, as a later block explains, the financial losses that delay causes are generally not what cargo insurance is written to pay.

4. A share in a maritime sacrifice. Under the seafaring principle called general average, a cargo owner can owe money for a rescue that never touched its own goods, because all cargo interests share the cost of a deliberate sacrifice made to save the voyage. The full mechanics belong to their own block below, but the possibility belongs on the identification list, because it surprises traders more than any other.

5. Storage-phase risks. Goods do not cease to be exposed when they stop moving: warehouse fire, water damage, and handling accidents in storage claim their share of losses.

The identification step is done route by route and cargo by cargo: the risks facing frozen fish in a reefer container are not the risks facing bagged cement.

KEY CONCEPT

Cargo insurance, the parties, and the policy's core numbers

Cargo insurance is insurance designed to provide financial protection against specified loss of or damage to goods during insured transportation or related activities. Its importance in logistics is

structural: international trade would be a parade of intolerable gambles if every loss fell in full on whoever happened to hold the goods at the moment of the accident. Insurance converts a ruinous, random loss into a known, budgetable cost, which is what allows businesses to ship at a scale no single loss could fund alone.

The parties are named plainly. The insured is the person or organisation covered under an insurance policy. The insurer is the company or organisation providing insurance coverage under an agreed policy. The deal between them is recorded in the policy, and three numbers in it matter most at this stage.

1. The insurance premium is the amount paid for insurance coverage. It is the price of transferring the financial consequence of the risk.

2. The sum insured is the monetary amount for which an insured interest is covered under the policy. It is the ceiling of the cover, and choosing it wisely is a discipline of its own, treated at the end of this lesson.

3. The deductible, met properly in the block on exclusions and conditions, is the portion of an insured loss borne by the insured.

The subject matter of cargo insurance is the goods themselves while they are in transit, together with the incidental exposures around the transit: loading, discharge, and interim storage connected with the journey. Cover can be arranged for the legs the parties actually face, including the whole door-to-door journey where the shipment is arranged so; a trader who ships constantly may cover movement after movement under a continuing arrangement rather than buying a separate policy for every consignment. What the policy protects is money: compensation to the insured within the agreed terms.

KEY CONCEPT

Insurable interest: who may insure

Not everyone with an opinion about a shipment may insure it. The governing principle is insurable interest: the insured must stand to gain from the cargo's safe arrival or suffer from its loss. The principle ties insurance cover to a genuine stake in the goods, and it exists to keep insurance a protection against real exposure rather than a wager on other people's misfortune.

Insurable interest follows the risk. The party who bears the risk of transit under the trade term has the clearest stake, because that party pays if the goods vanish. A seller usually holds the interest while the goods remain at its risk before dispatch; a buyer holds it once risk has passed; and where the term obliges one party to insure for the benefit of the other, as some Incoterms rules do, the arrangement exists precisely so the cover is in force at the moment the risk crosses over, with the policy document made transferable to the party who will hold the interest during the voyage.

Two practical consequences deserve attention. First, because insurable interest follows the risk, the allocation exercise of Lesson 4.3 is also the insurance exercise: work out where risk passes, and you have worked out who can and should insure each leg. Second, the interest must be real. A business cannot sensibly insure goods it will never own and never lose; a buyer who has not yet taken the risk has nothing to insure, and a seller who has passed the risk has, in general, little left to protect beyond its prospect of payment.

This is why the professional question is never simply "is this shipment insured?" but "who holds the risk at this leg, and is the cover in that party's interest?" The next block meets a party whose responsibilities are real but which is not the answer to that question.

KEY CONCEPT

Carrier liability is not cargo insurance

Carriers also answer for cargo, and the two channels must never be confused. Carrier liability is the legal or contractual responsibility of a carrier for specified cargo loss, damage, or delay. When goods are damaged in a carrier's custody, the carrier may indeed be answerable. But that responsibility is neither automatic nor unlimited.

Carriage is governed by applicable transport rules and by the carrier's own contract of carriage, and across the international carriage regimes two features recur. First, carriers enjoy defences: there are circumstances, such as damage arising from certain inherent characteristics of the goods or from events beyond the carrier's control, in which the carrier is not answerable at all. Second, liability is limited: where a carrier is answerable, the rules and the contract commonly cap the compensation well below the commercial value of the goods, and they attach their own notice requirements and time limits to claims. The specific figures differ by regime and are not taught in this course; the structural point is what matters.

Cargo insurance behaves differently in every particular. It is a separate contract, bought in advance by the cargo interest, that responds to covered losses within its own terms, and it does not depend on proving the carrier at fault. The two channels also answer different questions: the carrier's responsibility asks what the carrier owes for what happened on board; the policy asks what the insurer owes the insured under the agreed cover.

The practical consequences are uncomfortable but clarifying. An uninsured cargo owner whose goods are destroyed may finish the episode with a fraction of the value recovered even where the carrier was plainly implicated, because defences and limits stand in the way. Conversely, holding insurance does not erase the carrier's responsibility. A professional treats them as two distinct recoveries, each with its own basis, its own documents, and its own clock.

KEY CONCEPT

The coverage ladder: Institute Cargo Clauses (A), (B), and (C)

Cargo cover is not one product but a ladder of three standard levels, the Institute Cargo Clauses, the marine-cargo policy wordings in general use across the international market, in the version

current since 2009. The levels are known by the letters (A), (B), and (C), and the alphabet climbs with the protection.

At the bottom stands Institute Cargo Clauses (C), which covers major casualties only: the listed catastrophes of the journey, among them fire and explosion, the vessel being stranded, grounded, sunk, or capsized, collision, derailment, or overturning of the conveyance, discharge of cargo at a port of distress, and the sacrifice of cargo under general average. At the middle level, Institute Cargo Clauses (B) is a named-perils cover that adds further listed events relative to (C), such as cargo washed overboard, entry of water into the conveyance or container, and loss or damage during loading and discharge. Named perils coverage is insurance covering losses caused by specifically identified risks or events: if the event is not on the list, the loss is not covered, however real it was.

At the top stands Institute Cargo Clauses (A), the all-risks level. All risks coverage is insurance covering fortuitous physical loss or damage except where excluded by the policy terms. The wording reverses the logic: instead of listing what is covered, it covers everything fortuitous and physical, and pushes the list in the other direction, to what is not covered. Even so it is not a blank cheque, and the exclusions get a block of their own below.

The premium rises with the ladder: (C) is the cheapest, (A) the dearest, which is why the choice is an economic decision and not an automatic one. The common market pattern follows the cargo: bulk, low-risk commodities, whose dominant dangers are the vessel's own casualties, often travel at the (C) or (B) level, while finished, high-value, damage-prone goods typically justify the wider protection of (A). Match the ladder to the exposure, and pay for breadth only where the cargo needs it.

KEY CONCEPT

Exclusions, policy conditions, and the deductible

An insurance exclusion is a circumstance, cause, or category of loss that is not covered by an insurance policy. Exclusions are not fine-print spite; they are the boundary lines that keep the cover insurable, and a professional reads them before the loss, not after.

Even the widest level, Institute Cargo Clauses (A), carries exclusions. In general terms, the standard clauses do not respond to loss caused by the wilful misconduct of the insured, to ordinary wear and tear and leakage, to inherent vice, to delay, and to war and strikes unless that cover is separately bought. Three of these deserve unpacking.

1. Inherent vice means deterioration arising from the goods' own nature: grain that heats in a hold, fruit that ripens and rots, moisture-laden goods that sweat and mildew. The insurer accepts the voyage; it does not accept the cargo's chemistry.

2. Delay excludes the financial consequences of lateness: the lost market, the stopped production line. This is the boundary line behind the delay family identified at the start of the lesson.

3. War and strikes risks sit outside the standard cover but can generally be bought back as separate extensions where the route requires them.

Policy conditions are the insured's own obligations under the agreement: to observe the duties the policy spells out, to notify claims promptly, and to allow the insurer's examination of the loss. A policy is a two-sided contract, and conditions breached at the wrong moment can prejudice a claim however genuine the loss.

The deductible is the portion of an insured loss borne by the insured under the applicable policy. Deductibles keep small, routine losses inside the trader's own accounts, and they mean that a claim is measured net of the agreed first slice. Their size varies by policy; this course teaches the concept, never amounts. Read the whole document: cover, exclusions, conditions, deductible.

KEY CONCEPT

Incoterms 2020: where risk passes and why that decides who insures

Incoterms 2020 is the International Chamber of Commerce's set of eleven three-letter rules that say who does what, who pays what, and where risk passes in an international sale; they do not decide who owns the goods or how payment is made. Lesson 4.3, Incoterms and Allocation of Responsibilities, taught the full allocation; this lesson needs the risk-transfer points, because they decide where insurable interest settles.

The risk passes from seller to buyer as follows. Under EXW, when the goods are placed at the buyer's disposal at the seller's premises. Under FCA, when the goods are handed to the buyer's carrier at the named place. Under FAS, when the goods are placed alongside the ship at the port of shipment. Under FOB, CFR, and CIF, when the goods are on board the vessel at the port of shipment. Under CPT and CIP, on handover to the first carrier. Under DAP, when the goods are placed at the buyer's disposal, ready for unloading, at the named destination; under DPU, the same but after unloading; and under DDP, as DAP with import clearance added for the seller's account.

One structural fact dominates the insurance question. Under the four C rules, CFR, CIF, CPT, and CIP, the seller pays for carriage to the destination, but the risk passes to the buyer much earlier, already at shipment. Paying for carriage does not mean bearing the risk to destination. A buyer purchasing on CFR or CIF therefore owns the sea voyage's risk while the seller owns the freight bill, and the two must never be confused.

The insurance conclusion follows mechanically from the sequence taught at the start of this lesson: identify the risk, allocate it by the term, then ask who insures. The party holding the risk between the transfer points is the party whose money is exposed, and that party is the natural buyer of cover for the leg. The next block applies this to the terms one by one.

KEY CONCEPT

Who arranges the policy under each trade term

Apply the sequence and the insurance map of Incoterms 2020 falls out quickly. Only two of the eleven rules contain a duty to insure. Under CIF, cost, insurance and freight, the seller must insure at minimum Institute Cargo Clauses (C) cover for the buyer's benefit; under CIP, carriage and insurance paid to, the seller must insure at the higher Institute Cargo Clauses (A) level, reflecting that CIP is used across all modes and commonly for higher-value, damage-prone consignments. Under CIF the insurance covers, at minimum, the shipment-leg risk from the port of shipment to the named destination port; under CIP, from the place of shipment to the named destination. A seller arranging either should make sure the insurance period runs over the full transit risk the buyer takes over, not merely the port-to-port sliver of it.

Under every other rule there is no obligation to insure at all, though the risk may make it wise. The buyer choosing EXW, FCA, FAS, or FOB carries the main carriage risk and insures, or gambles, as it sees fit. The seller choosing DAP, DPU, or DDP carries the risk to destination and is the natural insurer of the journey it owns. Under CFR the seller pays the freight but the buyer owns the voyage risk, so the buyer is the party who should hold a policy.

Three refinements matter in practice. First, the Incoterms insurance duty is a minimum, not a maximum: parties may agree more, a wider ladder level or a higher sum, and many do. Second, the customary valuation basis in the market, reflected in the clauses themselves, is to insure for 110 per cent of the contract value; the figure is a long-standing benchmark, and the parties' agreement governs. Third, the buyer under CIF or CIP needs proof of the cover promised: the certificate of insurance, the short document evidencing that a shipment is covered under the seller's marine policy, commonly required under CIF and CIP so the buyer receives transferable evidence of cover. It belongs to the certificate family met in Lesson 6.3, Certificates of Origin, Permits, and Supporting Documents: another document that proves, on its face, that something required is in place.

KEY CONCEPT

General average: the sacrifice every cargo shares

General average is the oldest sharing rule in shipping, and the one no trader should first meet in a demand for money. Codified for the modern era in the York-Antwerp Rules, first settled in 1890 and revised since, the principle is this: where a voluntary sacrifice, such as cargo jettisoned to lighten a stranded vessel, or an extraordinary expense, is deliberately made to preserve the common adventure from a common peril, the sacrifice is shared proportionally among the ship, the cargo, and the freight interests that the rescue saved.

Notice what the principle does to ordinary intuition. The sacrifice was made for everyone: the water that drowned one hold's cargo saved the hull, the remaining cargo, and the voyage's earnings. The law therefore spreads the cost over all who benefited, not merely over the unlucky. Every cargo interest contributes roughly in proportion to the value its cargo brought

safe to the adventure, and that includes cargo owners whose goods were never touched, never wetted, and never threatened directly. Cargo owners ordinarily post security for their contribution, typically through their insurers, before the goods are released for delivery.

Two connections bind this block to the rest of the lesson. First, general average sacrifice is such a recognised maritime peril that it appears among the major casualties covered even at the narrowest level, Institute Cargo Clauses (C); the ladder protects the contribution as well as the loss. Second, general average is why cargo insurance matters even to the trader whose container always arrives fine: an insured cargo interest hands the problem of security and contribution to its insurer, while an uninsured one must find its share from its own pocket before its goods move.

Real declarations of general average in recent years have included the Ever Given (2021) and Maersk Honam (2018) cases, and the first worked example of this lesson walks through such an event step by step, in words, from a cargo owner's chair.

KEY CONCEPT

When loss happens: the claim, the survey, and the documents

An insurance claim is a formal request for compensation under an insurance policy, and a claim well run is a procedure, not an outburst. The broad sequence runs as follows.

1. Notify promptly. The insured tells its insurer, and the carrier where the goods travelled in its custody, as soon as loss or damage is discovered. Prompt notice protects evidence, preserves the insured's rights against the carrier, whose liability rules attach their own time requirements, and satisfies the policy's conditions, which demand timely notice.

2. Limit the further loss. The insured takes sensible steps to prevent the damage spreading: segregating wet cartons, moving cargo out of the rain, salvaging what can be salvaged. The policy expects mitigation, not passivity.

3. Survey. The insurer arranges or requires an examination of the damaged cargo. The output is the survey report: a document recording the findings of an examination of cargo condition, loss, or damage. It establishes what happened, how extensive the loss is, and what caused it, and it anchors the claim's quantum.

4. Assemble the documents. A cargo claim is proved on paper. The documents commonly required include the insurance policy or certificate of insurance, the commercial invoice, the packing list, the transport document such as the bill of lading or air waybill, the survey report, and the correspondence with the carrier asserting its responsibility. They are the same document families whose verification Lesson 6.3, Certificates of Origin, Permits, and Supporting Documents, and the documentation lessons around it trained; here they serve a second master, the claim file.

5. Claim and settle. With notice given, survey done, and documents assembled, the insurer assesses the loss within the policy's terms, applies the deductible and any conditions, and compensates within the sum insured.

The habit to carry forward is simple: the claim begins on the day the damage is discovered, and its strength is decided by the promptness of notice and the quality of the file.

KEY CONCEPT

Insuring value: invoice value, landed cost, and under-insurance

One decision remains in the sequence this lesson opened with: insure, yes, but for how much? The tempting answer is the invoice value of the goods, because that is the number on the paper in front of everyone. It is usually the wrong answer, for a reason this course has already priced.

Lesson 10.2, Freight Costing and Landed-Cost Calculations, built the landed cost: goods value, freight, duty and taxes, destination charges, inland transport, and the tail. Every line of that build is money the buyer has genuinely at stake in the goods arriving. A consignment lost is not merely an invoice evaporating; it is the freight paid or payable, the duties and taxes incurred, the destination charges and inland carriage that will be re-incurred, and the selling position that depended on the goods. Under-insurance is the gap that opens when the sum insured is set on the invoice value while the true financial exposure runs to the full landed cost: whatever the loss, the compensation is bounded by the sum insured and by the policy's terms, so the uninsured slice of the exposure falls on the business that underinsured. Whether any particular cost line beyond the invoice is recoverable depends on the policy's wording, which is one more reason to read it before buying.

The professional disciplines follow. First, value the cover on the full exposure, using the landed-cost build as the map of what is at stake, and remember the market's customary benchmark of 110 per cent of contract value in CIF and CIP sales. Second, revisit the basis when costs move; a sum insured set two seasons ago may no longer bound today's exposure. Third, treat prevention as the first line of defence. Risk mitigation, the actions taken to reduce the likelihood or consequences of an identified risk, is cheaper than any claim: sound packaging, careful stowage, vetted carriers, sensible routes and seasons, secure storage, and accurate documentation all shrink the loss that no policy fully restores. Insurance stands behind what prevention cannot stop; prevention keeps the claim, the deductible, the delay, and the lost customer off the books in the first place.

WORKED EXAMPLE

General average aboard a container vessel

The following scenario is illustrative and told in words; it follows the general average principle as taught in this lesson. Step 1: the voyage. A container vessel is mid-Atlantic on a rotation from Shanghai toward West Africa. Among some thousands of containers are two belonging to

Adomah Foods Ltd, a processor in Accra, carrying food-processing machinery invoiced at 180,000 US dollars, sold on FCA Shanghai terms. All figures are illustrative for teaching purposes.

Step 2: the peril. A fire breaks out in a lower hold. The master faces a choice: lose the ship, or fight the fire in a way that destroys some cargo to save the rest. The master orders the hold sealed and flooded with seawater. The fire dies. Some containers in the hold are burnt beyond saving, dozens more are soaked, and the great majority of cargo on board, including both of Adomah Foods' containers, is untouched.

Step 3: the diversion. The vessel diverts to a nearby port for emergency repairs and re-inspection, an extraordinary expense incurred to preserve the voyage.

Step 4: the declaration. The vessel's owners declare general average. Specialists appointed to adjust it measure the values at stake and work out each interest's share of the sacrifice and the expense, roughly in proportion to the saved value each interest contributed to the adventure. Adomah Foods reads the notice with disbelief: its containers were never touched, yet before its machinery can be delivered, it must post security for its contribution to a rescue its goods did not need.

Step 5: the insurance difference. Because Adomah Foods had arranged all-risks cover as the FCA buyer, it hands the notice and the security demand to its insurer, which provides the security and absorbs the contribution within the policy terms. A neighbouring consignee on the same vessel, uninsured, must fund its share from working capital and wait, uninsured, through the adjustment.

What does this example demonstrate? General average spreads a voluntary sacrifice over every cargo interest that benefited, including the untouched, and security is ordinarily required before delivery. Insurance is the difference between forwarding the problem and funding it.

Key lesson: even a perfectly safe cargo is exposed on a shared voyage; insure the interest, not just the container.

WORKED EXAMPLE

Choosing cover: cocoa butter on CIP against clinker on CFR

Step 1: the two desks. In Tema, an exporter of cocoa products is booking 800 cartons of cocoa butter, a finished, high-value food product, to a buyer in Rotterdam, sold on CIP Rotterdam terms. Across the harbour office, a second trader is fixing a full bulk cargo of clinker, the low-value mineral input for cement, to a buyer overseas, sold on CFR terms. Both desks must now answer the same question: who insures, and at which level of the ladder? All figures and volumes are illustrative.

Step 2: the CIP desk applies the rule. CIP carries a duty to insure: the seller must arrange cover for the buyer's benefit at Institute Cargo Clauses (A), the all-risks level, with the insurance period running over the transit risk that passed to the buyer when the goods were handed to the first

carrier. The desk checks that the cover is real, obtains the certificate of insurance for the buyer, and notes that the customary market basis is 110 per cent of the contract value.

Step 3: the CIP desk reasons about the level. Even without the obligation, the level would choose itself: cocoa butter is finished, expensive, and damage-prone, exposed to heat, taint, and water, and the losses that matter to it, water entry and handling damage, sit above the narrow (C) rung. The mandatory (A) matches the cargo.

Step 4: the CFR desk applies the rule. CFR carries a freight obligation and no insurance duty. Risk passed to the buyer when the clinker went on board at the load port, so the buyer owns the sea voyage; the seller's stake in the transit is gone. Nobody is obliged to insure, but the buyer would be wise to, in its own interest.

Step 5: the CFR desk reasons about the level. If the buyer insures, the reasoning runs the other way: bulk clinker is low-value, robust, and its dominant dangers are the vessel's own casualties, fire, grounding, sinking, the very events of the (C) rung. A (C) or (B) cover is the natural consideration, and the premium economy is exactly why the ladder exists. The buyer that insures nothing is gambling the whole cargo value on the ship's luck.

What does this example demonstrate? The trade term allocates the risk, and the allocation decides who insures; the cargo's character decides the level. Same ladder, different seats.

Key lesson: first ask where risk passes, then ask who holds the exposure, and only then choose the rung.

WORKED EXAMPLE

The under-insurance gap: insured at invoice, exposed at landed cost

Step 1: the purchase. Sika Health Ltd, a pharmaceutical distributor in Accra, imports a consignment of medicines from a supplier overseas. The invoice value of the goods is GHS 400,000. Sika's clerk, filling in the proposal form, writes the sum insured as GHS 400,000, the number on the invoice. All figures in this example are illustrative for teaching purposes.

Step 2: what the consignment really cost. Recall the landed-cost build from Lesson 10.2. For this shipment, illustratively: ocean freight GHS 30,000; import duty, VAT, and levies computed by rule at the time of assessment, GHS 70,000; terminal and destination charges GHS 10,000; clearance fees and inland transport GHS 18,000. The full exposure bound up in this consignment is therefore about GHS 528,000, not the GHS 400,000 on the invoice. The duty and taxes alone represent money already spent or committed because of these goods.

Step 3: the loss. Mid-voyage, fire breaks out in the hold and Sika's consignment is destroyed. Fire is a major casualty, covered even at the narrowest level of the ladder, so the level of cover is not the problem here. The problem is the ceiling.

Step 4: the shortfall. The claim is measured within the policy's terms, and the sum insured the policy records is GHS 400,000. The gap between what the consignment cost Sika in total and what the sum insured can respond with, about GHS 128,000 in this illustration, falls on Sika. Whether particular cost lines beyond the invoice, such as the duty already paid, would have been recoverable had the cover been written more widely depends on the policy wording; the sum insured at invoice value guarantees only that the answer starts too low.

Step 5: the correction. On the next import, Sika values the cover from the landed-cost build: goods, freight, duties and taxes, destination charges, and inland carriage, with the market's customary 110 per cent of contract value in mind where the sale terms carry an insurance obligation. The premium is modestly higher; the exposure is finally matched.

What does this example demonstrate? The invoice is the price of the goods; the exposure is the landed cost. Insuring the first while risking the second is how businesses fund their own shortfalls.

Key lesson: set the sum insured from the build, not from the invoice.

IMPORTANT NOTE

Insurance pays money; it does not deliver the goods

The deepest misconception this lesson dismantles is the belief that insurance puts the transaction right. Insurance compensates: it moves money, within the policy's terms, after the loss. It does not deliver the goods, and it does not repair the damage the loss does to the business around the goods.

Consider what actually follows a serious loss. The customer who was to receive the consignment is still unserved, and the relationship is strained whether or not a claim is eventually paid. The production line waiting on the lost spare parts is still stopped while the survey, the file, and the settlement take their course. The trader must still arrange replacement supply, renegotiate delivery, and manage the commercial consequences, none of which the policy performs. And some of the damage is financial in a form the standard cover was never written to answer: the losses caused by delay, the missed market, the expired season, are excluded even under the widest level of the clauses.

Insurance is therefore a balance-sheet protection, not an operations plan. It preserves the money, and preserving the money is essential; but the service reputation, the customer's schedule, and the trading relationship are protected by the things insurance is not: prevention, buffer stock, reliable carriers, and honest communication when a shipment goes wrong. A professional buys the cover for the loss that would otherwise be unbearable, and still works as if the cover were not there.

IMPORTANT NOTE

"All risks" is not all events, and the minimum is not a maximum

Two boundaries close this lesson, and both are lines people cross in the wrong direction in real claims.

First, all-risks cover does not mean all events. Institute Cargo Clauses (A) covers fortuitous physical loss or damage except where the policy excludes it, and the exclusions are real: inherent vice, the goods' own nature working their deterioration; delay and the financial losses it causes; ordinary wear and leakage; wilful misconduct; war and strikes unless separately bought. A consignment of timber that seasons and splits in a humid hold has not suffered an insured event merely because the policy was titled all risks. The professional discipline is to read the exclusion list against the cargo's actual vulnerabilities before the voyage, and to buy back what the route genuinely needs as separate cover where that is available.

Second, the Incoterms insurance duty is a floor, not a ceiling. Under CIF the seller's obligation is minimum cover at Institute Cargo Clauses (C); under CIP, at the (A) level. Parties may agree more, and the minimum binds only where the contract is silent. Equally, under every other rule nobody is obliged to insure at all, which is not the same as saying nobody should. The term tells you the floor and the risk tells you the need; neither substitutes for reading the two documents that actually govern the money: the policy, with its sum insured, conditions, and exclusions, and the contract of sale, with the trade term and any agreed extensions written into it.

Practice questions (Lesson Checks)

Practice Question 1

A Ghanaian importer buys a consignment of machine parts on CFR Tema terms. During the ocean voyage, heavy weather damages several of the importer's containers. The importer demands that the seller compensate the loss, arguing that the seller paid the freight all the way to Tema.

Which statement best describes the position?

1. The seller must compensate, because paying the freight to the destination port means bearing the risk of the voyage to that port.

2. Under CFR the seller pays carriage to the destination, but the risk passed to the buyer when the goods were on board the vessel at the port of shipment, so the buyer bears the voyage loss and any insurance was the buyer's own arrangement. ✓

3. The shipping line automatically compensates the buyer for the full invoice value of the cargo, so neither trading party bears the loss.

4. The seller was obliged to insure the goods for the voyage under CFR, so the seller's insurer must pay the buyer's claim.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

This is the classic trap of the C rules. Under CFR, CIF, CPT, and CIP the seller pays for carriage to the destination, but the risk passes to the buyer already at shipment, when the goods are on board the vessel at the port of shipment. Paying for carriage does not mean bearing the risk to destination. The lesson's sequence applies: the trade term allocated the voyage risk to the buyer at shipment, so the buyer was the party with the insurable interest at sea and the party whose decision it was to insure or gamble. Only CIF and CIP oblige the seller to insure, at minimum ICC(C) and ICC(A) respectively; CFR contains no insurance duty at all.

Answer 1 restates the exact misconception: freight paid is a cost, not a risk allocation. Answer 3 overstates carrier liability, which is subject to the carriage rules' defences and limits and is never an automatic full-value guarantee. Answer 4 invents an insurance obligation that CFR does not contain.

The takeaway: read the term for two separate allocations, cost and risk; they pass at different points, and insurance follows the risk.

Practice Question 2

A container vessel suffers a serious fire mid-voyage. The master orders the burning hold flooded with seawater, saving the ship and most of the cargo. On arrival, an importer whose containers were completely untouched by the fire and the water is required to post security for a general average contribution before its goods can be delivered.

Which statement best explains the demand?

1. The demand is an error, because only the cargo owners whose goods were actually sacrificed or damaged can be asked to contribute.

2. The carrier has absorbed the loss as an ordinary operating cost of the voyage, and the contribution is simply the recovery of that cost.

3. A voluntary sacrifice made to preserve the common adventure from a common peril is shared proportionally among ship, cargo, and freight interests, including cargo owners whose goods were untouched, and insurers commonly provide the security before delivery. ✓

4. General average contributions arise only when the vessel is totally lost, so a demand after a successful salvage must be a scam.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

General average is the maritime principle, codified in the York-Antwerp Rules, under which a voluntary sacrifice or extraordinary expense deliberately made to preserve the common adventure from a common peril is shared proportionally among the ship, the cargo, and the freight interests that benefited. The flooding of the hold saved every cargo owner's voyage, so every cargo interest contributes roughly in proportion to the value its cargo brought safe to the adventure, and cargo owners ordinarily post security for their contribution, typically through their insurers, before delivery. This is exactly why the lesson teaches that cargo insurance matters even when the goods arrive fine: the insured hands the security and contribution problem to its insurer.

Answer 1 reflects the intuitive but wrong view; the untouched cargo owner owes a share precisely because the rescue was made for all. Answer 2 mischaracterises the sharing rule as an ordinary carrier cost. Answer 4 confuses general average with total loss; the principle operates precisely when the adventure is saved, not when it is lost.

The takeaway: on a shared voyage, a rescue for one is a charge on all, and insurance is how an untouched cargo owner pays without bleeding.

Practice Question 3

After a storm, a trainee finds cartons of imported stationery soaked and damaged. The carrier indicates that its responsibility for the goods is limited under the applicable carriage rules. The trainee's manager replies that the company holds an all-risks cargo policy, "so none of the carrier's position matters and everything will be paid regardless."

Which option best describes the correct view of the two channels?

1. The manager is right: an all-risks policy covers every loss from every cause, so the carrier's position is irrelevant once cover exists.

2. The carrier is automatically liable in full for all cargo damage occurring in its custody, so insurance is redundant whenever a carrier can be identified.

3. The insured must choose one channel only, and abandoning any claim against the carrier increases the amount the insurer must pay.

4. Carrier liability and cargo insurance are separate channels with different bases: the carrier's responsibility depends on the carriage rules and contract, with their defences and limits, while the policy responds only within its own terms, which exclude some causes of loss. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

Carrier liability is the legal or contractual responsibility of a carrier for specified cargo loss, damage, or delay, and it operates under the applicable carriage rules and the contract of carriage, which attach defences, limits, and time requirements to it. Cargo insurance is a separate contract that responds to covered losses within its own terms, without requiring the insured to prove the carrier at fault. The two channels are distinct in basis, documents, and timing, and a professional pursues each on its own merits rather than treating one as a substitute for the other.

Answer 1 is the manager's error: all-risks cover means all risks of fortuitous physical loss or damage except the exclusions, so causes such as inherent vice or delay sit outside it, and no policy makes the carrier's position irrelevant. Answer 2 overstates the carrier's responsibility, which is neither automatic nor unlimited. Answer 3 invents a forced election; nothing in the lesson's claim procedure requires abandoning one channel to use the other, and the claim procedure requires notifying both insurer and carrier promptly.

The takeaway: know what each channel can and cannot answer, and never let the comfort of a policy replace the discipline of reading its terms.

Fact-check appendix (audit notes — not for entry)

Insurance definitions (cargo insurance, insured, insurer, insurance premium, sum insured, insurance exclusion, deductible, all risks coverage, named perils coverage, carrier liability, risk transfer, insurance claim, survey report, risk mitigation) follow the lesson brief's key terms, defined at first use and used naturally in the blocks. — source: lesson brief 11.1, Key Terms (curriculum database record for slug cargo-insurance-and-risk-allocation; the brief file was absent from curriculum/lesson_briefs/module-11/ and the record was read from local_data/db.sqlite3).

Institute Cargo Clauses (A), (B), (C) ladder: (A) covers all risks of loss or damage subject to exclusions (wilful misconduct, ordinary wear/leakage, inherent vice, delay, war and strikes unless separately bought); (B) and (C) are named-perils covers; (B) adds washing overboard, water entry, and loss/damage in loading/discharge relative to (C); (C) covers major casualties only (fire, explosion, vessel stranding/grounding/sinking/capsizing, collision/derailment/overturning of conveyance, discharge at a port of distress, general average sacrifice); premium rises with the ladder, C cheapest, A dearest; standard market wording current version 2009. — source: research sheet trade-transport-standards.md, section 10.

Typical uses of the ladder (bulk, low-risk commodities at (C)/(B); finished, high-value, damage-prone goods at (A)) are taught as the common market pattern at the level of the research sheet and the lesson assignment, without any figures. — source: general business knowledge, consistent with research sheet trade-transport-standards.md, section 10.

General average: voluntary sacrifice or extraordinary expense to preserve the common adventure from a common peril, shared proportionally among ship, cargo, and freight interests; codified in the York-Antwerp Rules (first 1890, updated 1994/2004/2016); cargo owners post security, typically via insurers, before delivery; modern declarations include the Ever Given (2021) and Maersk Honam (2018) cases. The worked-example fire scenario is a labelled illustrative story, not attached to any named real vessel. — source: research sheet trade-transport-standards.md, section 10.

Insurable interest: the insured must stand to gain from the cargo's safe arrival or suffer from its loss. — source: research sheet trade-transport-standards.md, section 10.

Certificate of insurance: the short document evidencing that a shipment is covered under an open/marine policy, commonly required under CIF/CIP so the buyer receives assignable evidence of cover. — source: research sheet trade-transport-standards.md, section 10.

Incoterms 2020 insurance duties: only CIF and CIP oblige the seller to insure for the buyer's benefit, CIF at minimum Institute Cargo Clauses (C), CIP at the higher (A) level; cover at minimum the shipment-leg risk from the port/place of shipment to the named destination; sellers should ensure the insurance period covers the full transit risk passed to the buyer; the customary market basis is 110 per cent of the contract value, taught as a long-standing benchmark reflected in the clauses, not as an exact universal rule; the duty is a minimum and parties may agree more. — source: research sheet trade-transport-standards.md, sections 1 and 10, including safe phrasings and DO NOT CLAIM item 10.

Incoterms 2020 risk-transfer points taught: EXW at the seller's premises; FCA on handover to the buyer's carrier; FAS alongside ship; FOB and CFR/CIF when goods are on board the vessel at the port of shipment; CPT/CIP on handover to the first carrier; DAP ready for unloading at destination; DPU after unloading; DDP as DAP plus import clearance; the four C rules pass risk at shipment while the seller pays carriage to destination; Incoterms do not govern ownership or payment. — source: research sheet trade-transport-standards.md, section 1.

Carrier liability: taught as the concept (legal or contractual responsibility for specified cargo loss, damage, or delay, subject under carriage regimes to defences, limits, and time/notice requirements) without any monetary limits, deadlines, or regime names with figures; this course teaches no liability figures. — source: general business knowledge within the lesson brief's key-term definition; WRITING_SPEC.md rule 5.1 (no amounts, deadlines, or free periods stated as fact).

Claims procedure (notify promptly, mitigate, survey, assemble documents, settle within terms) and the commonly required document list (policy or certificate, commercial invoice, packing list, transport document, survey report, carrier correspondence) are taught generically without deadlines or amounts. — source: general business knowledge, consistent with research sheet trade-transport-standards.md, section 10 (survey and certificate concepts) and the lesson brief's objectives.

Landed cost as the measure of the buyer's full exposure (goods, freight, duties and taxes, destination charges, inland transport, tail) and unit-vs-invoice contrast follow the costing lens of Lesson 10.2, referenced by name only. — source: curriculum draft lesson-10.2 (Freight Costing and Landed-Cost Calculations); WRITING_SPEC.md section 7.

Cross-references by name only, without re-teaching: Lesson 4.3 Incoterms and Allocation of Responsibilities (risk transfer); Lesson 6.3 Certificates of Origin, Permits, and Supporting Documents (document families, one line each); Lesson 10.2 Freight Costing and Landed-Cost Calculations (landed cost). — source: WRITING_SPEC.md section 7; lesson titles from the curriculum database.

All companies (Adomah Foods Ltd, Sika Health Ltd), goods, quantities, invoice values, and every GHS and US dollar figure in the worked examples are illustrative for teaching purposes only and represent no actual premium, sum insured, tariff, tax, or levy amount. No premium levels, deductible amounts, claim deadlines, or percentages beyond the sheet's customary 110 per cent benchmark appear anywhere in the lesson. — source: WRITING_SPEC.md fact-safety rule 5.1; research sheet DO NOT CLAIM items 5 and 10.

Lesson 11.2 — Dangerous Goods and Special Cargo Handling

Summary

This lesson examines the handling and transportation requirements for dangerous goods and cargo requiring specialised care. Learners will explore dangerous goods classifications, hazard communication, packaging, labelling, documentation, and applicable transport regulations. The lesson also introduces refrigerated, oversized, fragile, and high-value cargo, highlighting the importance of qualified personnel, appropriate equipment, and compliance with relevant safety requirements.

Learning objectives

- Define dangerous goods and special cargo.
- Identify the major hazard classes used in dangerous goods transportation.
- Explain the purpose of dangerous goods classification, packaging, marking, and labelling.
- Recognise the roles of applicable maritime, air, and land transport regulations.
- Identify the documentation commonly required for dangerous goods shipments.
- Describe basic handling considerations for temperature-sensitive, oversized, and fragile cargo.
- Explain the importance of trained personnel and approved handling procedures.
- Recognise the consequences of incorrect dangerous goods declarations.
- Identify situations requiring specialist advice or authorisation before transportation.

Key terms

Dangerous Goods — Substances or articles presenting hazards during transportation and regulated under applicable transport safety requirements.

Hazard Class — A category used to classify dangerous goods according to their primary transport hazards.

UN Number — A four-digit identification number assigned to specified dangerous goods or groups of dangerous goods under the United Nations system.

Proper Shipping Name — The designated transport name used to identify a dangerous substance or article under applicable dangerous goods regulations.

Safety Data Sheet (SDS) — A document providing information about a chemical's hazards, safe handling, storage, and emergency measures.

Dangerous Goods Declaration — A document or electronic declaration providing required information about dangerous goods offered for transportation.

IMDG Code — The International Maritime Dangerous Goods Code governing the maritime transport of dangerous goods.

IATA Dangerous Goods Regulations (DGR) — Industry regulations published by IATA for air transport of dangerous goods, based on applicable international requirements.

ICAO Technical Instructions — International technical instructions governing the safe transport of dangerous goods by air.

Refrigerated Cargo — Goods requiring controlled temperatures during transportation or storage.

Out-of-Gauge Cargo (OOG) — Cargo whose dimensions exceed the standard internal or external limits of the intended transport equipment.

Fragile Cargo — Goods requiring protection against damage from shock, vibration, pressure, or improper handling.

Hazard Label — A label communicating the hazard classification of dangerous goods.

Segregation — The separation of incompatible dangerous goods or cargo according to applicable safety requirements.

Teaching content

KEY CONCEPT

Two kinds of cargo that change the rules

Earlier courses treated cargo largely as a neutral quantity: cartons, bags, drums, and containers whose management questions were questions of cost, space, and time. Module 11 turns to risk and compliance, and this lesson deals with the cargo that changes the rules completely, because of what it can do when something goes wrong.

Dangerous goods are substances or articles presenting hazards during transportation and regulated under applicable transport safety requirements. Flammable liquids, compressed gases, corrosive chemicals, and lithium batteries are ordinary commercial examples. Two features of the definition deserve attention. First, the test is hazard in transport: a product may be safe on a shop shelf, yet present a genuine hazard inside a closed ship's hold, an aircraft hold at altitude, or a steel container heating up on a tropical road. Second, the regulation is not optional: if goods meet the definition, the requirements apply whoever ships them, and however small the consignment.

Special cargo is cargo requiring additional handling, documentation, packaging, or transportation precautions because of its characteristics. Frozen fish, temperature-sensitive medicines, machinery wider than a container opening, glass products, live animals, and high-value consignments all sit in this family. Dangerous goods are one branch of special cargo, and the most strictly regulated one, which is why they take the larger half of this lesson.

Why is compliance treated as absolute, when so much else in logistics is negotiated? Because the scale of failure is different. An ordinary cargo failure is a claim: goods are damaged or lost, an insurer pays, and the loss is measured in money. A dangerous-goods failure is a catastrophe: a fire in a vessel's hold far from land, a toxic release in a terminal, an event in an aircraft hold from which there is no recovery. Catastrophes are not compensated afterwards; they are prevented beforehand, which is why dangerous-goods rules are written as duties rather than choices, and why the first of those duties, a true declaration of what the goods are, sits with the shipper.

This lesson builds in four stages: the classification system, the compliance chain from packaging to acceptance, the modal rulebooks and the people who operate them, and the wider families of special cargo.

KEY CONCEPT

Classification is the foundation: the safety data sheet and the shipper's duty

Every control in dangerous-goods compliance depends on getting one prior step right: classification, the process of deciding what the goods are under the system the rulebooks share. If classification is wrong, everything built on it is wrong: the packaging is selected for the wrong hazard, the label announces the wrong danger, the segregation decision is made on false information, and the emergency services plan for an event that will not be the one that happens. Classification is therefore not clerical work at the end of an export job. It is the foundation on which the whole consignment's compliance stands.

Where does a shipper start? With the safety data sheet (SDS), a document providing information about a chemical's hazards, safe handling, storage, and emergency measures. Manufacturers issue the SDS in a standard sixteen-section format, and it is the starting point for deciding whether goods are regulated for transport at all, and under what entry. Lesson 7.2, Air Cargo Operations and Cargo Acceptance, worked through reading an SDS for an air shipment; the same discipline opens every dangerous-goods movement in every mode.

Classification produces three outputs, and each becomes a permanent feature of the shipment.

1. The hazard class. A hazard class is a category used to classify dangerous goods according to their primary transport hazards. Classification answers which of the nine classes described in this lesson the goods belong to.

2. The UN number. A UN number is a four-digit identification number assigned to specified dangerous goods or groups of dangerous goods under the United Nations system, written with the prefix UN. UN 1950 identifies aerosols and UN 3480 identifies lithium-ion batteries, to take two widely met examples.

3. The proper shipping name. The proper shipping name is the designated transport name used to identify a dangerous substance or article under the applicable dangerous goods regulations.

Trade names and marketing descriptions are not transport names; the proper shipping name is the one the rulebooks recognise.

The same chemical carries the same UN number in every country and every rulebook; what changes between modes is not the classification but the procedures built around it. And the duty is personal: the shipper owns classification. A supplier's brochure or a customer's assurance does not discharge it. The shipper reads the SDS, applies the rulebook, and stands behind the result.

KEY CONCEPT

The nine hazard classes

The United Nations Model Regulations, on which all the modal rulebooks are built, divide dangerous goods into nine classes. Lesson 3.3, Cargo Types, Packaging, and Handling, introduced them briefly within its cargo map; this lesson treats them as a working system, because a logistics professional must be able to place real products into real classes.

1. Class 1, Explosives, divided into divisions 1.1 to 1.6 according to the nature of the explosive risk.

2. Class 2, Gases, with three divisions: 2.1 flammable gases, 2.2 non-flammable non-toxic gases, and 2.3 toxic gases.

3. Class 3, Flammable liquids, the class behind countless everyday products, from solvents and paints to fuels.

4. Class 4, Flammable solids, with three divisions: 4.1 flammable solids and self-reactive substances, 4.2 substances liable to spontaneous combustion, and 4.3 substances which emit flammable gas in contact with water.

5. Class 5, Oxidising substances and organic peroxides, with 5.1 oxidising substances and 5.2 organic peroxides.

6. Class 6, Toxic and infectious substances, with 6.1 toxic substances and 6.2 infectious substances.

7. Class 7, Radioactive material, a class of its own with no divisions.

8. Class 8, Corrosive substances, which attack living tissue and metal.

9. Class 9, Miscellaneous dangerous substances and articles, the class that gathers hazards not fully covered elsewhere. It matters increasingly in ordinary logistics, because it includes lithium batteries, dry ice, and environmentally hazardous substances.

Three observations complete the picture. Divisions within a class distinguish different forms of one primary hazard, which is why gases divide by flammability and toxicity. Class 9 shows that

the system moves with commerce: a consignment of electronics containing lithium batteries, or a frozen consignment chilled with dry ice, is a dangerous-goods consignment even though it looks harmless. And classification turns on the primary hazard in transport, which is exactly the judgement the safety data sheet supports.

KEY CONCEPT

Packing groups: degree of danger, not size of box

Within most classes, the rulebooks grade how dangerous one substance is compared with another of the same class. The instrument is the packing group. Packing group I denotes great danger, packing group II medium danger, and packing group III minor danger. Two flammable liquids in the same class can present very different levels of danger, and the packing group records that difference on the documents and drives the strictness of everything downstream.

Two boundaries matter. First, packing groups do not apply to every class: they are assigned in most classes but not in classes 1, 2 and 7. Explosives, gases, and radioactive material are governed by their own class-specific requirements instead. Second, and this is the error to dismantle now: a packing group is a degree of danger, not a packaging size. It does not mean a large drum, a medium box, or a small bottle. It measures how dangerous the substance is, and that measurement then determines the performance the packaging must deliver. The link between the two is performance packaging, treated in the next block, but the group itself is a statement about the substance, never about the box.

Reading packing groups correctly matters operationally: the group appears on the dangerous goods declaration, drives packaging selection, and informs stowage and segregation decisions. Misreading it as a size instruction produces absurd documents; ignoring it removes the grading that tells handlers which of two flammable liquids deserves the greater care.

KEY CONCEPT

Compatibility and segregation

Some dangerous goods must not travel together. Segregation is the separation of incompatible dangerous goods or cargo according to applicable safety requirements, and it exists because some hazards interact: combined, they produce a risk that neither poses alone. The rulebooks carry the detailed requirements in tables that trained professionals consult; at the level of this lesson, the professional needs the concept and the habit.

The concept is clearest at the classic pairing. A flammable liquid supplies the fuel for a fire; an oxidising substance supplies the oxidiser that feeds it. Held apart, each is a managed hazard. Stowed together in an enclosed space, each makes the other's failure worse, which is why such combinations are kept apart by the segregation rules. Between that extreme and ordinary compatible stowage lies a spectrum of restrictions, and the rulebook table, not intuition, decides which applies to any given pairing.

The habit is procedural, and it connects directly to operations this programme has already taught. Lesson 5.3, Shipping Instructions and Cargo Consolidation, taught the checks a forwarder performs before consolidating several shippers' goods into one container. Lesson 7.3, Containerisation: FCL, LCL, and Container Types, taught how containers are packed. Wherever cargo from different sources comes together, in a container freight station or a consolidation, segregation becomes a live question, and the check is documentary: what classes are present, are they compatible, and does the rulebook allow this combination in this configuration? The check costs minutes. The failure it prevents can cost a vessel.

KEY CONCEPT

Packaging, marking, and labelling: making the package speak

Classification identifies the hazard; the next links in the compliance chain contain it and communicate it. Each link exists for a purpose the professional should be able to state, not merely recite.

1. Performance packaging. Dangerous goods travel in UN-performance packaging: packaging designed, tested, and marked to perform to a standard matched to the hazard, rather than ordinary trade packaging chosen for appearance or price. The purpose is containment that survives the drops, stacking, vibration, and moisture of a real journey, and the packing group grades how demanding that standard must be.

2. Marking. Packages are marked with the proper shipping name and the UN number. The purpose is identification: anyone handling the package can know what it is without opening a file. A package marked only with a trade name is anonymous to the stevedore, the inspector, and the firefighter alike.

3. Hazard labels. A hazard label is a label communicating the hazard classification of dangerous goods, applied to the package in the standardised form the rulebooks prescribe. The purpose is instant recognition of the danger: the label speaks to a handler who has no time and no documents, and it informs the stowage, segregation, and emergency-response decisions taken at every interchange.

4. Placards. Where the mode requires them, placards, larger signs serving the same purpose as labels, are fixed to transport units and vehicles, so that the hazard of a whole load is visible from outside, to inspectors and, if the worst happens, to emergency services deciding how to respond.

The common idea is that the cargo announces itself. A dangerous-goods consignment correctly packaged, marked, and labelled carries its own safety instructions on its face, and every participant in the chain, including ones the shipper will never meet, can act on what they see.

KEY CONCEPT

The dangerous goods declaration and the shipment's document set

A dangerous goods declaration is a document or electronic declaration providing required information about dangerous goods offered for transportation. It is the shipper's declaration: made in the form the applicable rulebook prescribes, given under the shipper's own authority, and stating the facts the rulebook requires. For a typical consignment those facts are the ones classification produced: the UN number, the proper shipping name, the hazard class, the packing group where one applies, the quantities and type of packaging, and confirmation that the goods are correctly packaged, marked, and labelled as the requirements demand.

The declaration's significance is easiest to see by imagining its absence. The carrier would be accepting cargo whose nature it does not know; the vessel or aircraft would be stowed without regard to the hazard; the crew would respond to an incident blind; and every downstream control would be built on silence. The declaration is the moment the shipper's knowledge becomes the whole chain's knowledge. That is why it is the shipper who gives it, and why a false or careless declaration is treated as one of the most serious failures in logistics.

Around the declaration sits the shipment's wider document set. The transport document itself, the bill of lading for a sea consignment or the air waybill for an air one, carries the dangerous-goods entries, so that the cargo's nature appears on the documents the whole trade runs on. The safety data sheet remains available through the information chain, so that handlers and responders can consult the full hazard picture. And the carrier's booking and manifest carry the dangerous-goods details, so that the vessel's or aircraft's stowage can be planned before the cargo moves. A dangerous-goods consignment is therefore documented into the declaration, into the transport documents, and into the carrier's records, and the entries must agree.

KEY CONCEPT

One classification, three rulebooks: IMDG, IATA DGR, and the road question

The United Nations system supplies the classification, and each transport mode applies it through its own rulebook. The professional must know which book governs which journey.

1. Sea: the IMDG Code. The IMDG Code, the International Maritime Dangerous Goods Code, is the rulebook governing the maritime transport of dangerous goods, mandatory under Chapter VII of the SOLAS convention and updated on a two-year amendment cycle. A container shipped from Tema towards Rotterdam travels under the IMDG Code for the sea leg.

2. Air: the ICAO Technical Instructions and the IATA DGR. The ICAO Technical Instructions are the international technical instructions governing the safe transport of dangerous goods by air. The IATA Dangerous Goods Regulations (DGR) are the industry regulations published by IATA for the air transport of dangerous goods, based on the ICAO Technical Instructions and updated annually. Lesson 7.2 taught acceptance under the DGR; the discipline is at its strictest in this mode, for the reason this lesson opened with: an aircraft cannot stop.

3. Road: the special case of ADR. ADR is a European agreement on the road transport of dangerous goods. It is the source of a persistent error, so the boundary must be stated plainly: ADR does not apply in Ghana or elsewhere in West Africa. What travels globally is the classification system, the nine classes and the UN numbers, which ADR shares with the IMDG Code and the DGR. The road leg of a West African journey follows the applicable national rules, and a professional who writes European references onto a West African road consignment has decorated the file rather than complied with anything.

For a multimodal movement the classification is constant while the regime switches leg by leg: the same drum of paint is a class 3 consignment under the IMDG Code aboard the vessel, under national road rules on the truck, and under the IATA DGR if it flies. Decide the mode first, then prepare the documents from that mode's rulebook.

KEY CONCEPT

Trained people, acceptance checks, and the price of a wrong declaration

Every rulebook described so far assumes one thing the documents cannot supply: competent people. Dangerous-goods rules work only in the hands of personnel trained for their role, which is why requirements for training and authorisation are built into the rulebooks themselves, and why Lesson 7.2's air acceptance discipline, dangerous goods accepted only by trained and authorised staff checking the consignment against the regulatory checklist, is the visible tip of a rule that runs through every mode. The sea terminal's checker, the vessel's officer planning stowage, the driver of a tank truck, and the warehouse clerk signing off a consolidation each hold a role the system depends on.

The acceptance check is where the compliance chain closes. Before a carrier, a terminal, or a consolidator takes a dangerous-goods consignment, trained staff verify that the classification is plausible, the packaging and marks are correct, the labels are present, and the declaration is complete. Lesson 7.2 taught this as air cargo acceptance; the same gate exists on the sea side, to catch failures that slip past even conscientious shippers.

When the declaration is wrong, whether carelessly or deliberately, the protections do not degrade gracefully; they disappear. Packaging selected for the wrong hazard fails, stowage ignores the real danger, segregation is computed from false premises, and the crew that faces a fire or a leak reads documents describing a different cargo. Beyond the physical risk sits the commercial and legal one: consignments refused or quarantined, penalties imposed, liability for the consequences of the misdeclaration, and the loss of the standing that allows a shipper to move cargo at all. An incorrect declaration is not a paperwork error with a paperwork consequence. It is the removal of every protection the system built, committed to paper by the one participant who knew the truth.

KEY CONCEPT

Refrigerated cargo and the documented cold chain

The second half of this lesson's title belongs to cargo that is not dangerous but still punishes negligence. Refrigerated cargo is goods requiring controlled temperatures during transportation or storage, and its compliance discipline is documentary: the cold chain must be provable, not merely maintained.

Lesson 7.3 treated the refrigerated container as equipment: insulated, fitted with its own cooling unit, and demanding electrical power wherever it stands. The compliance view adds four questions.

1. What is the set point, and who says so? The required temperature comes from the product's storage requirements, confirmed in writing at booking, not from the driver's habit or the previous shipment's setting. A wrong set point, faithfully maintained, still delivers spoiled cargo.

2. Was everything pre-cooled? The reefer container is pre-cooled before loading, and cargo whose condition depends on temperature is brought to the required temperature before it enters the box. Loading warm goods into a cold container shocks the goods and spends the equipment's capacity recovering instead of holding.

3. Does power continue at every leg? The container needs electricity aboard the vessel, in the terminal, and on the road, where a generator set supplies it. Every change of leg is a handover of responsibility for power, and a silent break in supply can ruin cargo that still looks normal on arrival.

4. Where are the records? Temperature recorders travel with the cargo, and the equipment logs its own performance. Downloaded at destination, these records are the journey's evidence.

That evidence is what decides disputes. When a temperature excursion, a period spent outside the required range, occurs and a claim follows, the temperature record locates the failure in time: which leg, whose custody, whose responsibility. A party with records can show where the chain broke; a party without records argues. Pharmaceutical shipments make the discipline strictest: medicines require documented temperature control from end to end, with the documentation treated as part of the product's quality, because a medicine that has travelled outside its range may be worthless even when it looks perfect.

KEY CONCEPT

Out-of-gauge, fragile, and high-value cargo: when specialist advice comes first

Out-of-gauge cargo (OOG) is cargo whose dimensions exceed the standard internal or external limits of the intended transport equipment. Lesson 7.3 met its equipment: open-top containers for over-height pieces, flat racks for machinery that no closed box can swallow. The compliance questions ride on top of the equipment choice. An out-of-gauge lift is engineered before it is performed: a lifting plan sets the method, the equipment, and the rigging before a crane ever

takes the load, because improvising with a piece weighing many tonnes is how terminals have accidents. An out-of-gauge road move is surveyed before it is driven: a route survey confirms that the roads, bridges, bends, and overhead clearances on the chosen path can actually pass the load. Both are concepts every planner should recognise, and both are work for specialists.

Fragile cargo is goods requiring protection against damage from shock, vibration, pressure, or improper handling. Its handling considerations are ordinary in kind and unforgiving in degree: packaging that cushions and supports, orientation marks that are respected, loading plans that keep weight off delicate pieces, and handling by people and equipment briefed on what is inside the crates.

Two further families complete the map, treated here lightly. Live animals travel under welfare requirements set by regulation: containers designed for the animals' needs, care arrangements for the journey, and the permissions the movement requires, all arranged before departure. High-value cargo attracts security requirements: sealed and controlled handling, restricted access, and sometimes authorised or escorted procedures, so that the cargo's value does not become its vulnerability.

The unifying rule of the special-cargo half of this lesson is this: the moment a consignment's characteristics exceed the ordinary operator's competence or equipment, specialist advice or authorisation must come before transportation, not after a failure. A chemical never shipped before, a lift beyond the terminal's usual capacity, a route nobody has surveyed, a document set that does not clearly fit any pattern: each is a signal to stop and consult. The cheap failure is the question asked early.

WORKED EXAMPLE

Classifying and shipping a cleaning compound from Tema

A trading company in Tema supplies industrial cleaning products to an overseas buyer, shipping by sea from Tema Port. A new formulation is ordered in drums, and the company's export clerk must take it through the full dangerous-goods path for the first time. The product and details in this example are illustrative for teaching purposes.

Step 1: Read the safety data sheet. The clerk obtains the manufacturer's SDS. Its transport-relevant sections identify the product as a flammable liquid graded packing group II. Under this lesson's classification, that makes it a class 3 consignment: the classification the SDS supports, never the one the trade name suggests.

Step 2: Identify the UN entry. From the SDS and the IMDG Code's index, the clerk confirms the proper shipping name and the UN number under which the product travels. The trade name remains on the commercial invoice; the proper shipping name goes onto the marks and the declaration.

Step 3: Select performance packaging. The product is packed in UN-performance packaging appropriate to a packing group II flammable liquid: drums tested and marked to the standard, not reconditioned containers of unknown history.

Step 4: Mark and label. Each drum is marked with the proper shipping name and the UN number, and carries the class 3 hazard label, applied so that it survives the journey and stays legible.

Step 5: Declare. The clerk completes the dangerous goods declaration for the sea leg under the IMDG Code, stating the UN number, proper shipping name, class, packing group, quantities, and packaging, and books the consignment with the shipping line as dangerous goods, so that the vessel's stowage can be planned for it.

Step 6: Acceptance. At the terminal, trained staff check the consignment before loading: marks, labels, declaration, packaging. The check passes, the drums are stowed in line with the vessel's dangerous-goods arrangements, and the shipment sails.

What does this example demonstrate? The compliance chain is a sequence in which each step consumes the previous one: classification feeds the packaging decision, the packaging feeds the marking, the marks feed the declaration, and the declaration feeds the acceptance check. Skipping a step does not speed the shipment; it unbuilds every step after it.

Key lesson: the shipper who walks the chain in order owns a compliant consignment. The declaration is credible precisely because every step behind it was actually performed.

WORKED EXAMPLE

The consolidation check that caught an incompatible pair

A freight forwarder consolidating groupage cargo at a container freight station near Tema Port is filling a forty-foot container for a regional sea service. Two consignments are on the receiving list: household goods, and a consignment of swimming-pool treatment chemicals documented as class 5.1, oxidising substances. A third booking arrives late: a consignment of paint documented as class 3, flammable liquids. Space is tight, and the warehouse supervisor's first plan slots the paint into the same container to fill the last gap. This example is illustrative for teaching purposes.

Step 1: The check. Lesson 5.3 taught that consolidation is a documented checking process, not a loading exercise. Among the checks is compatibility: what classes are present in this container, and are they permitted together? The forwarder's DG-trained clerk pulls both declarations before approving the plan.

Step 2: The finding. An oxidising substance and a flammable liquid are the classic incompatible pairing: one supplies the fuel for a fire, the other supplies the oxidiser that feeds it. Sealed together in a container at sea, each makes the other's failure worse. This is precisely the

combination the segregation concept exists to question, and the clerk will not approve it on personal judgment alone.

Step 3: The decision. The paint is booked onto the next sailing instead, into a container carrying no oxidising cargo. The rework costs the forwarder an illustrative additional handling charge, a day of replanning, and a difficult telephone call to the paint consignor, whose vessel connection slips by a week.

Step 4: The counterfactual. Had the check been skipped, the container would have sailed with the two consignments stowed together. In all probability nothing would have happened: most voyages pass without incident, which is exactly why incompatible consolidations accumulate, because the absence of disaster reads as proof of safety. But given a leaking paint drum, a failed valve on a chemical drum, and heat in the hold, the vessel's crew would have been fighting a fire fed by its own cargo, with documents that listed each consignment separately and no combination anyone had approved.

What does this example demonstrate? Segregation is a check with a place in the process, and the place is before the container is packed, while the plan is still on paper and the fix costs a telephone call rather than a salvage operation.

Key lesson: the consolidation check that feels redundant on a quiet afternoon is the one standing between routine cargo and a stowage decision nobody would defend afterwards.

WORKED EXAMPLE

A cold chain documented end to end: pharmaceuticals from Rotterdam to Accra

A pharmaceutical distributor in Accra imports a temperature-sensitive medicine from a manufacturer whose regional distribution centre stands in Rotterdam. The product must be held within a specified temperature range until it is delivered in Accra, and both parties treat the documentation of that range as part of the product itself. The product, route, and details in this example are illustrative for teaching purposes.

Step 1: The set point is fixed in writing. At booking, the required temperature range is stated on the booking and confirmed against the product's storage requirements. The reefer container is specified, and the set point is recorded on the equipment handover documents.

Step 2: Pre-cooling and loading. At Rotterdam, the container is pre-cooled to its set point before stuffing, and the medicine, already held at the required temperature, is loaded. An independent temperature recorder is placed with the cargo, logging the journey separately from the container's own equipment.

Step 3: Power continuity at every leg. The container travels to the port on the terminal's power, crosses to Tema aboard the vessel on the ship's power supply, and completes the road leg to Accra on a generator set. At each interchange, the power source and the set point are verified and recorded, because each is a change of custody as well as a change of plug.

Step 4: The excursion. At Accra, the distributor downloads the temperature records before releasing the medicine for sale. The record shows one excursion: a rise above the required range lasting some hours during the road leg, after the container left the terminal.

Step 5: The claim, decided on records. The distributor rejects the affected stock and claims. The records decide the question cleanly: the container's log and the cargo recorder agree that temperature held correctly through the sea voyage and the terminal stay, and the excursion began only on the road leg, in the haulier's custody. The haulier's generator log for the same period confirms a power interruption. Responsibility locates itself in the documents: the claim is pursued against the party whose custody matches the failure, and the earlier, unaffected legs are not dragged into it.

What does this example demonstrate? The cold chain was not merely maintained; it was evidenced. Records converted what could have been a three-way argument into an evidence question with one answer.

Key lesson: in temperature-controlled logistics, the documentation is not paperwork about the cargo. It is the proof of what the cargo experienced, and it is worth exactly as much as the discipline with which each leg was recorded.

IMPORTANT NOTE

"It is only a small quantity" is how dangerous-goods incidents begin

Every dangerous-goods trainer meets the sentence: it is only a small quantity, a few bottles, one carton, a sample for the customer. The sentence is the most dangerous in this subject, and this lesson ends its authority here.

The rules do not regulate danger by the pallet. They regulate substances and articles that present hazards in transport, and a single leaking bottle of flammable liquid in a closed container, an undeclared aerosol in an aircraft hold, or one damaged lithium battery in a carton of mixed goods can begin an event that no quantity of paperwork afterwards will stop. Quantity changes the value of the goods; it does not change the nature of the hazard they can start.

Undeclared dangerous goods are the gravest compliance failure in this subject, graver than a late declaration, a wrong label, or a packaging defect, because they strip every protection silently. Lesson 7.2 taught why air acceptance checks consignments rather than trusting them: the system assumes that some shippers will not declare, and builds a last line of defence at acceptance. A professional who finds regulated goods travelling as general cargo stops the consignment, whatever the cost of stopping it.

The working rule is symmetrical. If the goods are regulated, declare them, whatever the quantity. If you are not willing to declare them, do not ship them.

IMPORTANT NOTE

Packing groups grade danger, not packaging size

The most common beginner's misreading of the classification system is to hear packing group and picture packaging: group I, a big strong crate; group III, a small light carton. The picture is wrong, and documents built on it are wrong with it.

A packing group is a statement about the substance: how much danger it presents within its class. Packing group I denotes great danger, II medium danger, III minor danger. What the group does downstream is select the performance standard the packaging must meet, so that a greater danger travels in packaging tested to a more demanding standard. The size and shape of that packaging follow the product and the journey, not the group.

Remember also the boundary taught earlier: packing groups are not assigned at all in classes 1, 2 and 7. Explosives, gases, and radioactive material carry no packing group on their documents, and their packaging requirements come from their own class rules. A document showing a packing group against a gas cylinder is not a variant reading. It is an error to correct before the consignment moves.

IMPORTANT NOTE

The SDS starts the process but never replaces the declaration

Two errors live at the boundary between the safety data sheet and the dangerous goods declaration, and this note closes both.

The first is omission: a shipper reads the SDS, decides the product looks harmless, and ships it as general cargo. The rule this lesson teaches is that if the SDS says regulated, the goods are regulated. The shipper does not get a second opinion on the hazard. The shipper owns the classification, and optimism is not a classification method.

The second is substitution: a shipper attaches the SDS to the consignment and considers the dangerous-goods obligation discharged. It is not. The SDS is the starting point for deciding whether goods are regulated for transport, and under what entry; it is not the transport declaration. The declaration is a separate document, in the form the modal rulebook prescribes, given by the shipper, carrying the UN number, proper shipping name, class, packing group, and packaging confirmation. The SDS informs the classification; the declaration declares it. A consignment with an SDS and no declaration is a consignment whose hazards are documented and whose compliance does not exist.

The professional habit that joins the three stages: SDS first, classification second, declaration third, and nothing shipped until all three agree.

Practice questions (Lesson Checks)

Practice Question 1

A trading company in Accra is shipping twelve bottles of a flammable cleaning solvent by sea to an overseas buyer. The product's safety data sheet identifies it as a flammable liquid. The sales manager proposes declaring it as general cargo, arguing that twelve bottles is too small a quantity to matter.

Which statement best describes the correct course of action?

1. The consignment may ship as general cargo, because quantities this small fall outside the dangerous-goods classification system.

2. The consignment must be classified, packaged, marked, labelled, and declared as dangerous goods, because regulation follows the hazard identified in the safety data sheet, not the quantity offered for shipment. ✓

3. The consignment may ship as general cargo by sea, although a declaration would be required if the same goods travelled by air.

4. The freight forwarder may decide whether a declaration is required, since the forwarder accepts the consignment on the carrier's behalf.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

Dangerous goods are regulated because of the hazard they present in transport, and the safety data sheet is the starting point for deciding whether goods are regulated. A flammable liquid does not become unregulated because the consignment is small: quantity changes the value of the cargo, not the nature of the hazard. The compliant path is the full chain taught in this lesson: classify from the SDS, use performance packaging, mark and label the packages, and make the dangerous goods declaration under the rulebook governing the mode, which for this sea shipment is the IMDG Code.

Answer 1 is the exact misconception this lesson warns against: undeclared dangerous goods are the gravest compliance failure in the subject, because they silently remove packaging, stowage, segregation, and emergency-response protections. Answer 3 invents a sea exemption that does not exist; the sea leg has its own rulebook and its own declaration. Answer 4 misplaces the duty: the declaration is the shipper's own responsibility and cannot be delegated to the forwarder's discretion.

The takeaway: if the goods are regulated, declare them whatever the quantity; if you are not willing to declare them, do not ship them.

Practice Question 2

While preparing documentation for a consolidated shipment, a clerk enters a packing group against a consignment of compressed gas cylinders and explains to a colleague that packing group II means the cylinders must travel in a medium-sized package.

Which statement best describes the clerk's work?

1. Both the entry and the explanation are in order, because gases carry packing groups and packing group II fixes the size of package required.

2. The packing group entry is wrong, because packing groups are not assigned in classes 1, 2 and 7, and the explanation is wrong, because a packing group grades the degree of danger a substance presents, not the size of its packaging. ✓

3. The packing group entry is wrong because gases always carry packing group I, but the explanation about package size is correct.

4. Both are acceptable in practice, because packing groups are optional entries that shippers may apply as they prefer.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

The clerk has made both errors this lesson isolates. Compressed gas cylinders belong to class 2, and packing groups are assigned in most classes but not in classes 1, 2 and 7; explosives, gases, and radioactive material are governed by their own class-specific requirements, so no packing group belongs on the cylinders' documents. Even where a packing group does apply, it is a statement about the substance: packing group I denotes great danger, II medium danger, and III minor danger. The group determines the performance standard the packaging must meet; it says nothing about the size or shape of the package, which follows the product and the journey.

Answer 1 accepts both errors. Answer 3 corrects the entry with a fiction, since gases carry no packing group at all, while keeping the size misconception. Answer 4 treats a grading system as a free-form field, which it is not: the packing group appears on the dangerous goods declaration and drives packaging selection and segregation decisions.

The takeaway: the packing group answers how dangerous, never how big, and for classes 1, 2 and 7 the correct answer is that no packing group exists.

Practice Question 3

A freight forwarder in Accra arranges the final leg of a dangerous-goods consignment: a container discharged at Tema Port is trucked to a warehouse in Kumasi. A colleague preparing

the file states that the European ADR agreement governs this road leg, because ADR is the world rulebook for dangerous goods by road.

Which statement best describes the applicable position?

1. The colleague is right, because ADR applies wherever dangerous goods travel by road, including West Africa.

2. The colleague is wrong, because no rules govern dangerous goods on West African roads, so the consignment may be documented as ordinary cargo for the truck leg.

3. The colleague is wrong, because ADR is a European agreement that does not apply in West Africa; the nine classes and UN numbers travel globally, and the road leg follows the applicable national rules. ✓

4. The colleague is wrong, because the IMDG Code that governed the sea voyage continues to govern the goods until delivery, including the road leg.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The classification system is global: the same substance carries the same hazard class and UN number in every rulebook, which is why the consignment classified for the sea voyage under the IMDG Code needs no reclassification for the road leg. But the rulebooks themselves are modal and regional. ADR is a European agreement on the road transport of dangerous goods, and it does not apply in Ghana or elsewhere in West Africa. The road leg of this journey follows the applicable national rules. The professional error to avoid is decorating a West African road file with European references and mistaking that for compliance.

Answer 1 states the exact misconception this lesson dismantles. Answer 2 overcorrects into lawlessness: that ADR does not apply does not mean nothing applies, and documenting regulated goods as ordinary cargo is the gravest failure in the subject. Answer 4 confuses the classification, which does continue with the goods, with the rulebook, which changes with the mode: the IMDG Code governs the sea leg, not the truck.

The takeaway: the classification travels; the rulebook is chosen by the mode, and on West African roads the applicable rules are national.

Fact-check appendix (audit notes — not for entry)

The UN Model Regulations divide dangerous goods into nine classes with the names and divisions taught in this lesson: 1 Explosives (divisions 1.1–1.6); 2 Gases (2.1 flammable, 2.2 non-flammable non-toxic, 2.3 toxic); 3 Flammable liquids; 4 Flammable solids (4.1 flammable solids/self-reactive, 4.2 liable to spontaneous combustion, 4.3 emit flammable gas with water); 5 Oxidising substances and organic peroxides (5.1 oxidisers, 5.2 organic peroxides); 6 Toxic and infectious substances (6.1 toxic, 6.2 infectious); 7 Radioactive material; 8 Corrosive substances; 9

Miscellaneous dangerous substances and articles, including lithium batteries, dry ice, and environmentally hazardous substances — source: research sheet trade-transport-standards.md section 9

UN numbers are four-digit identifiers with the UN prefix; UN 1950 (aerosols) and UN 3480 (lithium-ion batteries) are the sheet's worked examples, and the upper bound of the UN-number range is not taught — source: research sheet section 9

Packing groups grade danger for packing performance: PG I great danger, PG II medium danger, PG III minor danger; they apply to most classes except classes 1, 2 and 7, which have none — source: research sheet section 9 and DO NOT CLAIM item 7

Modal rulebooks: the IMDG Code governs sea transport and is mandatory under SOLAS Chapter VII with a two-year amendment cycle; the IATA DGR is based on the ICAO Technical Instructions and is updated annually; ADR is a European UNECE agreement that does not apply in Ghana/West Africa, while the classification and UN-number system are global and West African road moves follow national rules — source: research sheet section 9 and DO NOT CLAIM item 8

The SDS is the sixteen-section GHS document from the manufacturer; it is the starting point for deciding whether goods are regulated for transport and under what UN entry, but the SDS itself is not the transport declaration — source: research sheet section 9

The dangerous goods declaration is the shipper's modal declaration stating UN number, proper shipping name, class, packing group, quantities and packing/labelling compliance — source: research sheet section 5

The shipper uses UN-performance packaging, marks packages with UN numbers and proper shipping names, applies hazard labels on packages and placards on units/vehicles as the mode requires — source: research sheet section 9

Dangerous goods are accepted for air transport only by trained, authorised staff checking against the IATA DGR checklists; this lesson generalises acceptance checking as practice across modes — source: research sheet section 8

Container types: reefer (refrigerated, insulated), open top (removable roof, crane-loaded over-height cargo), flat rack (out-of-gauge machinery), ISO tank (tank in a frame for liquids, often dangerous goods) — source: research sheet section 6

Definitions of dangerous goods, hazard class, UN number, proper shipping name, SDS, dangerous goods declaration, IMDG Code, IATA DGR, ICAO Technical Instructions, refrigerated cargo, out-of-gauge cargo, fragile cargo, hazard label, and segregation are used as given in the lesson brief's key terms; the special cargo definition follows the lesson 7.2 brief — source: curriculum lesson briefs module-11/lesson-11.2.md and module-07/lesson-7.2.md

Cold-chain practices (set point confirmed at booking, pre-cooling of container and cargo, generator-set power on road legs, temperature recorders downloaded at destination, records as claim evidence), project-cargo practices (lifting plans, route surveys), live-animal welfare permissions, and high-value security handling are taught generically without named standards or amounts — source: general logistics practice

No amounts, percentages, packing-instruction numbers, limited-quantities thresholds, or specific national regulation names are stated; all worked-example details are labelled illustrative per writing spec rule 5.1 — source: writing spec WRITING_SPEC.md section 5

Cross-references: Lesson 3.3 Cargo Types, Packaging, and Handling; Lesson 5.3 Shipping Instructions and Cargo Consolidation; Lesson 7.2 Air Cargo Operations and Cargo Acceptance; Lesson 7.3 Containerisation: FCL, LCL, and Container Types; next lesson 11.3 Supply Chain Security and Cargo Integrity — source: curriculum lesson briefs

Lesson 11.3 — Supply Chain Security and Cargo Integrity

Summary

This lesson introduces security risks affecting international logistics networks and the measures used to protect cargo, facilities, information, and transportation systems. Learners will examine cargo theft, tampering, unauthorised access, fraud, and shipment diversion. The lesson also explores container seals, access control, cargo inspections, traceability, and the importance of secure procedures and accurate records.

Learning objectives

- Define supply chain security and cargo integrity.
- Identify common security threats affecting logistics operations.
- Explain how cargo theft, tampering, and fraud affect supply chain performance.
- Describe the purpose of container seals and cargo security inspections.
- Identify common physical security measures used in logistics facilities.
- Explain the importance of access control and personnel identification.
- Describe how shipment tracking and audit trails support cargo security.
- Recognise the role of documentation and data security in protecting shipments.
- Explain how security incidents should be reported and investigated.
- Identify preventive measures that reduce cargo security risks.

Key terms

Supply Chain Security — The measures used to protect goods, people, facilities, information, and transport systems against security threats throughout the supply chain.

Cargo Integrity — The assurance that goods remain complete, authentic, and protected against unauthorised interference or alteration.

Cargo Theft — The unlawful taking of goods during transportation, handling, or storage.

Cargo Tampering — Unauthorised interference with cargo, packaging, seals, or transport equipment.

Container Seal — A device used to indicate whether a secured container access point has been opened or interfered with.

High-Security Seal — A seal designed to meet specified security and mechanical requirements under an applicable standard.

Access Control — Procedures and systems restricting entry to authorised persons or vehicles.

Chain of Custody — A documented record of the possession, control, handling, or transfer of goods or evidence.

Shipment Traceability — The ability to reconstruct a shipment's history, movement, and relevant handling events.

Security Inspection — An examination performed to identify potential security threats, tampering, or unauthorised items.

Cargo Diversion — The unauthorised or fraudulent redirection of goods from their intended movement or destination.

Audit Trail — A chronological record enabling transactions, movements, or changes to be traced.

Security Incident — An event that threatens or compromises the safety, integrity, or security of logistics operations.

Teaching content

KEY CONCEPT

What supply chain security and cargo integrity mean

Every international consignment carries an unwritten promise: the goods that leave the shipper's premises are the goods that arrive — the same goods, in the same quantity, in the same condition, with nothing added, nothing removed, and nothing altered on the way. Cargo integrity is the assurance that goods remain complete, authentic, and protected against unauthorised interference or alteration. It is the promise itself.

Supply chain security is the system that keeps the promise: the measures used to protect goods, people, facilities, information, and transport systems against security threats throughout the supply chain. The breadth matters: security covers not only cargo but the people who handle goods, the warehouses and terminals that hold them, the documents and data that describe them, and the vehicles, vessels, and aircraft that move them. A weakness in any one of these can undo a consignment whose physical handling was otherwise careful.

The distinction matters. Cargo integrity is the state we want the goods to be in; supply chain security is the set of measures that produces and proves that state. Security without integrity is paperwork; integrity without security is luck.

The second defining idea is shared responsibility. An international consignment typically passes through many hands: the shipper who packs it, the haulier who collects it, the warehouse or terminal that stores it, the carrier that moves it, the customs administrations that examine it, and the destination agents who deliver it. No single party can keep the promise alone: every handover is an opportunity for interference, and every party depends on the discipline of those before it. Security is therefore a property of the chain, not of any one link. This lesson covers that chain: its threats, controls, seals, records, and the habits that keep the promise verifiable.

KEY CONCEPT

The threat landscape: theft, tampering, diversion, and fraud

Module 11 treats risk through a repeating cycle: identify what could go wrong, assess how likely and how damaging it would be, mitigate what matters, and monitor the result. Security is one of the recognised risk categories in logistics, standing alongside supply-side, transport, customs, financial, and natural risks.

1. Theft and pilferage. Cargo theft is the unlawful taking of goods during transportation, handling, or storage. It ranges from pilferage, the removal of small quantities whose individual losses pass unnoticed, to the theft of whole vehicles or containers; pilferage is the quieter and more corrosive form, quantities drifting away until a reconciliation reveals a pattern.

2. Tampering and contamination. Cargo tampering is unauthorised interference with cargo, packaging, seals, or transport equipment. A tampered load may have been opened to remove part of its contents and resealed, or adulterated or contaminated.

3. Insertion of undeclared goods. Criminals sometimes place undeclared or prohibited goods inside a legitimate, properly declared load. For an innocent shipper this is the most dangerous category, because the container carries their name and their declaration. The declaration disciplines taught in the air cargo acceptance lesson (Lesson 7.2) and the dangerous goods lesson (Lesson 11.2) exist partly for this reason: you must be able to show that you knew, and controlled, everything packed into your consignment.

4. Cargo diversion. Cargo diversion is the unauthorised or fraudulent redirection of goods from their intended movement or destination — a load collected by someone posing as the buyer's driver, or rerouted to a destination that was never agreed.

5. Documentation and data fraud. False release instructions, forged identities, altered delivery details: documents and data are targets too, treated later in this lesson.

Security threats are not confined to crime against cargo. Attacks on shipping in the Red Sea from late 2023 pushed major container lines to reroute around the Cape of Good Hope — a reminder that security events can reshape whole transport systems, not only single shipments.

KEY CONCEPT

How security failures damage supply chain performance

A security event is never only a security event. Its consequences spread through the same performance measures the rest of this course prices and manages.

1. Direct loss. Goods stolen, contaminated, or diverted are usually gone in physical and commercial terms: stock unavailable to sell, materials unavailable to produce, capital written off.

2. Delay. An investigation stops a shipment. A container that has become evidence — seal checked, photographs taken, authorities notified — is not being unpacked and sold. Examinations, holds, and re-inspections consume days in the way delay consumes days everywhere else in logistics; the next lesson returns to managing those consequences.

3. Claims and insurance friction. The previous lesson, Cargo Insurance and Risk Allocation, covered how cover responds to loss. Claims move faster and settle more cleanly where records show reasonable care — seals checked, handovers signed, anomalies reported — and they stall where the records show silence.

4. Contractual damage. Deliveries that arrive short, late, or altered breach the commitments made to customers, damaging service performance and, over time, the commercial relationship itself.

5. Scrutiny. An operator associated with a security incident can expect closer examination from customs and from commercial partners afterwards. Risk-based selection systems remember.

6. Reputation. Buyers choose suppliers they can trust to keep the promise: a demonstrated security discipline is a selling point, a concealed incident the opposite.

The pattern across all six: the cost of the event itself is usually smaller than the cost of handling it badly. That is why the later blocks of this lesson put as much weight on records, reporting, and investigation as on fences and seals: prevention and discipline are the operation's insurance against an incident becoming a crisis.

KEY CONCEPT

Container seals and the seal discipline

A container seal is a device used to indicate whether a secured container access point has been opened or interfered with. The definition carries the key idea: a seal does not make a container unopenable — any seal can be broken. What it does is make unauthorised opening detectable, converting the question of whether anyone opened the box from a matter of trust into a matter of evidence.

Two properties make a seal useful as evidence. First, it is applied once, at packing, and breaking it leaves visible marks. Second, it carries a unique number. A high-security seal is a seal designed to meet specified security and mechanical requirements under an applicable standard; such seals combine physical strength with unique, recorded numbering, at little cost against the cargo they protect.

The seal discipline has four movements.

1. Apply and record at packing. The shipper seals the container when packing is complete, and the seal number goes onto the shipping documents — on ocean shipments, the bill of lading carries the seal number alongside the container number.

2. Verify at every handover. Each party receiving the container checks the seal against the documents before accepting it: the number must match, and the seal must be intact. The equipment interchange receipt introduced in Lesson 9.3 records container and seal condition at each interchange between parties.

3. Protect continuity in transit. Customs transit regimes work on the same logic: goods moving in transit in West Africa travel under customs seals with a transit document, so the seal, rather than repeated inspection, vouches for the load at each border.

4. Treat a broken or mismatched seal as a stop event. A seal that is missing, broken, or numbered differently from the documents means the container's history cannot be vouched for. The response is to stop: do not open, do not repair, do not continue quietly — document and report, as the first worked example shows.

KEY CONCEPT

Security inspections

A security inspection is an examination performed to identify potential security threats, tampering, or unauthorised items. Inspections differ from quantity-and-condition checks in what they look for: not whether the goods are present and undamaged, but whether anything shows interference, and whether anything is present that should not be.

Inspections occur at predictable points in a consignment's life.

1. At packing and dispatch. Before the seal is applied, the packed goods are checked against the packing list. The close of packing is the moment the shipper certifies that it knows exactly what is inside.

2. At facility gates. Terminals, warehouses, and depots check vehicles and seals on entry and on exit.

3. At receipt. The receiving party checks seals and external condition before opening, then verifies contents against the documents.

4. At customs. Customs administrations examine selected consignments; in Ghana, the Customs Division of the Ghana Revenue Authority carries public security and safety among its responsibilities, alongside revenue and trade facilitation. Selection is risk-based: modern systems assess declarations and route them toward release without inspection, document check, or physical examination, and Ghana's ICUMS platform operates such risk management. A consignment linked to anomalies attracts examination more often — one more way poor security returns as delay.

Two disciplines make inspections worth their cost. First, inspect against a document, never against memory: a count compared with a packing list detects a shortfall, while a count compared with yesterday's impression detects nothing. Second, record the result: an inspection

that leaves no record cannot later protect the person who performed it, because nobody can prove it happened.

KEY CONCEPT

Physical security measures in logistics facilities

Facilities are where cargo is most concentrated and most accessible, so facility security is the foundation beneath every other control. The common physical measures form layers.

1. Perimeter protection. Fencing or walls around the site, with a controlled number of gates. A perimeter's purpose is not to make entry impossible but to channel every person and vehicle through points where they can be seen and checked.

2. Gates and vehicle control. A monitored gate, with vehicles registered on arrival, checked on exit, and documented when departing with cargo.

3. Lighting. Yards, loading bays, and storage approaches kept lit after dark. Darkness is cover; light is a cheap control.

4. Surveillance and alarms. Cameras covering entrances, loading areas, and high-value storage, with alarm systems actively monitored — equipment that records but is never reviewed is decoration.

5. Secure storage areas. High-value or sensitive goods held in locked cages, rooms, or segregated areas inside the general warehouse, under key control: limited keys, issued to named people, each issuance recorded.

6. Supervised loading and unloading. Loading bays staffed whenever cargo is moving; no cargo left unattended on a dock, and no driver left alone with an open load.

None of these measures is sophisticated, and that is the point: most integrity failures are opportunistic, and opportunity shrinks fastest under layers of simple, consistently applied controls. Facility security also earns its keep commercially — the premises, their access control, and their supervision are exactly what a trusted-operator assessment examines, as the third worked example shows.

KEY CONCEPT

Access control and personnel identification

Access control refers to the procedures and systems restricting entry to authorised persons or vehicles. Where physical security controls the ground, access control controls the people on it — and theft, tampering, and insertion all require access. Removing opportunity is the first preventive measure there is.

1. Identification for everyone on site. Staff carry photographic identification or badges; roles are visible; visitors and contractors sign a register, receive a visitor pass, and are escorted in operational areas.

2. Vehicles and drivers verified against documents. Before any load is released, the driver presenting the collection documents is checked against them: identity, vehicle registration, and the collection authorisation itself. The release discipline of Lesson 9.3 — cargo is released only against proper documents to the party entitled to them — is access control applied at the gate. A forged collection document defeated by a two-minute identity check is the cheapest fraud prevention in logistics.

3. Limits matched to roles. Access to high-value storage, to the seal store, and to document areas is restricted to the roles that need it.

4. Personnel screening over the employment life. Vetting at engagement for staff in sensitive positions — handling goods, seals, or data — and withdrawal of passes, keys, and system access the day a person leaves are part of the same control.

5. Records of entry and exit. Who entered, when, driving which vehicle, for which collection: the gate log is the facility's chain-of-custody record for people.

The principle beneath all five: integrity requires knowing who can touch the goods. An operator that cannot say who was on its premises at a given hour can neither defend its consignments nor hold the certification this lesson ends with.

KEY CONCEPT

Chain of custody, shipment traceability, and the audit trail

A chain of custody is a documented record of the possession, control, handling, or transfer of goods or evidence. Shipment traceability is the ability to reconstruct a shipment's history, movement, and relevant handling events; the audit trail, a chronological record enabling transactions, movements, or changes to be traced, is what makes it possible. The three interlock, compressing into a rule: at any moment, the records should answer who has it, and every change of answer evidenced.

Handovers are signed rather than waved because each signature — the haulier's at the warehouse gate, the terminal's at its out-gate, the driver's at receipt — is the moment responsibility moves: it ends one party's custody, begins the next party's, and fixes the condition and seal status at that instant. Module 8's record-keeping discipline applies: every movement leaves a record. For a containerised consignment, the chain-of-custody documents typically include the packing record and seal number at origin, the transport document, the equipment interchange receipt at each interchange, gate logs, handover signatures, and the delivery receipt, layered over scan and status events. Track-and-trace presents the trail as progress information; for security it is the investigation tool. In Ghana, ICUMS assigns every consignment a Unique Consignment Reference supporting end-to-end tracking.

The security value runs in both directions. After an event, the trail reconstructs the journey: where the consignment was, who held it, when seals were checked and numbers recorded, and where any anomaly first appears, as the second worked example shows. Before an event, gaps in the trail are warnings: a handover without a signature, a scan missing where every other consignment has one, a seal number absent from an interchange record: each a place where integrity runs on trust alone, an anomaly to fix, not a formality to skip.

Two habits keep the chain honest.

1. Known parties only. Custody transfers only to vetted carriers, drivers, and warehouses — checked and approved before the first handover, not discovered during it; each unknown party is a gap in the record and the assurance.

2. Sign only for what you verified. A signature attests the seal matched, the count checked, the condition as recorded; signing first and checking later converts the record from evidence into fiction. Record events as they happen, not afterwards from memory: a trail written the next day is a story; a trail written in the moment is evidence.

KEY CONCEPT

Documentation and data security

Cargo has a physical existence and an information existence, and the two can be attacked separately. A container behind a good fence is still lost if released against a forged instruction — goods surrendered to a false identity do not come back. Protecting shipments therefore means protecting their documents and data as deliberately as their steel and cartons.

The fraud vectors are mostly variations on impersonation: falsified release or delivery instructions collected under false identities; manipulated consignment details; documents altered after issuance; and declarations that misdescribe what a load contains.

The established controls are procedural rather than technical.

1. Document verification. Descriptions, quantities, and references checked for consistency across the document set before anything is acted on — the verification discipline earlier modules applied to customs documents applies here.

2. The three-way match. In commercial flows, the purchase order, the goods receipt, and the supplier invoice are matched before payment is released. The same triangulating logic applies to security: instruction, physical reality, and record must agree before goods move.

3. Approved-party lists. Collections, deliveries, and document handling are conducted only with carriers, brokers, and agents on an approved list — the documentary cousin of the known-parties rule in the chain-of-custody block.

4. Instruction verification. A change to delivery details or bank details received by email is verified by calling the counterparty on a known number before it is acted on. Call-back verification costs minutes; a diverted load costs the load.

5. Data access control. System accounts are individual, passwords protected, and access rights limited to role needs — the data-side twin of premises access control. A shared login cannot serve as an audit trail, because it identifies nobody.

Documentation fraud was named among the threats at the start of the lesson; these five controls are its answer.

KEY CONCEPT

Security incidents: reporting and investigation

A security incident is an event that threatens or compromises the safety, integrity, or security of logistics operations. The definition's breadth is deliberate: a broken seal, a missing pallet, a suspicious approach to a driver, an unexplained weight discrepancy — each is an incident whether or not goods were actually lost. Treating near-misses as incidents is itself a control, because the pattern of small events maps where the next loss will come from.

The response discipline has five movements.

1. Report promptly. Escalate internally at once, then to the carrier, the agent, and, where relevant, the police and customs. Prompt notification also matters commercially: insurance claims under the cover the previous lesson described expect notice without delay.

2. Preserve evidence. Do not repair, clean up, repack, or discard. Photograph seals, packaging, and surroundings; retain the broken seal and the documents; note times, names, and vehicle registrations while memory is fresh. The container that is evidence is inconvenient; the container that was tidied is worthless.

3. Reconstruct with the audit trail. The investigation traces the consignment through its records to locate the break point — the last moment the chain of custody was provably intact.

4. Establish root cause. The break point explains the mechanism; the root cause explains the failure that allowed it: a gate not logged, a handover unsigned, an unvetted subcontractor.

5. Correct and follow up. Fix the process, verify the fix, and feed the finding back into the risk cycle. An investigation that ends at blame has produced a scapegoat; one that ends at a changed process has produced security.

An incident hidden today is an incident discovered later, with the concealment attached: certification audits, claims investigations, and commercial disputes all read records, and records tell.

KEY CONCEPT

Trusted-trader programmes and the preventive habit set

The controls in this lesson are not only defensive. Internationally, they have become a qualification. Under the World Customs Organization's SAFE Framework of Standards, first adopted in 2005, customs administrations and businesses are encouraged into a partnership with a simple logic: an operator that can demonstrate sound security controls and a compliant record can be certified by its customs administration as a trusted trader — the concept known internationally as Authorised Economic Operator status — and receive facilitation in return: faster processing and fewer inspections. Mutual recognition arrangements between customs administrations extend that certification onto other trade lanes: an operator trusted at home is recognisable abroad.

A trusted-operator assessment looks at what this lesson has taught: the operator's premises and physical security; its documented procedures for handling cargo, seals, and documents; its personnel screening and access control; and its record-keeping — whether its audit trails genuinely show who handled what, when, and under which seal. The third worked example follows one exporter preparing for exactly this.

Why should a logistics professional care? Because security certification is increasingly a commercial qualification, not a bureaucracy: buyers and carriers ask for it, and an operator that holds it moves through the system more easily than one that must be inspected.

The lesson closes with the preventive habit set.

1. Verify what you receive: seals, counts, and condition, against documents.

2. Record what you hand over: signatures, seal numbers, scans, times.

3. Question anomalies: the mismatched seal, the weight that disagrees with the container's declared verified gross mass, the package that is not on the list. An anomaly questioned is a control; an anomaly explained away is a vulnerability.

4. Report promptly: incidents and near-misses alike.

WORKED EXAMPLE

The seal number that did not match: a delivery stopped at Tema

An importer of packaged consumer goods in Accra regularly receives containers discharged at the Port of Tema. This example reconstructs a delivery on which the seal discipline earned its keep; all details are illustrative for teaching purposes.

Step 1: Packing at origin. The exporter packs 900 cartons of assorted goods into a forty-foot container, seals the doors with a high-security seal, and records the seal number — for illustration, 4471902 — on the packing list and in the shipping instruction. The number appears on the bill of lading issued for the shipment.

Step 2: Arrival and haulage. The container is discharged at Tema. The importer's agent arranges haulage; at the terminal gate the container's condition and seal number are recorded on the equipment interchange receipt, as Lesson 9.3 described, and the driver departs for Accra.

Step 3: The check at delivery. At the importer's warehouse, the driver follows the routine this lesson builds: compare the seal against the documents before opening. The seal on the door reads 4472266. The bill of lading says 4471902. The numbers differ.

Step 4: The stop. The driver does not cut the seal and does not open the container. He photographs the seal and the document face, notes the time, and telephones the agent. The agent notifies the shipping line and the police. The container stands sealed in the yard.

Step 5: The aftermath. With the seal intact and the discrepancy photographed and documented, the carrier and the authorities can investigate a container whose evidence is preserved: gate records, interchange records, and the physical seal all speak. In the illustrative reconstruction, the container was found to have been opened after it left the terminal — and because the importer stopped at receipt instead of accepting and stripping quietly, the discovery remained attributable and traceable rather than becoming an unexplained shortfall absorbed into a stock count.

What does this example demonstrate? The seal discipline is not a formality at the end of a journey; it is the tripwire. One driver comparing one number against one document converted an unknown integrity failure into a documented, investigable event — and kept his employer from taking custody of a container whose history it could not vouch for.

Key lesson: check the seal before you open the box, and when the number disagrees, stop, document, and report. The consignment can wait; the evidence cannot.

WORKED EXAMPLE

Weight as evidence: reconstructing a shortfall in Singapore

A food importer in Singapore strips an import container and finds 540 cartons on the floor against 600 on the packing list. Sixty cartons — for illustration, of bottled sauce — are simply not there. All figures in this example are illustrative for teaching purposes.

Step 1: What the weight record says. The verified gross mass is the verified total gross mass of a packed container, determined by the shipper under the international safety requirements; a container without a verified gross mass may not be loaded, as Module 7's containerisation lesson explained. This container's declared figure was 21,400 kg; the terminal weighbridge at the load port read 21,430 kg. The yard record and the declaration agree.

Step 2: What the seal record says. The container was sealed at packing, and the seal number runs unchanged through every recorded handover: the bill of lading, the equipment interchange receipt at gate-out, and the delivery check at the importer's warehouse all carry the same number, and the seal at stripping matches.

Step 3: Reasoning from the records. Two hypotheses could explain a shortfall: the goods never left the origin warehouse in full, or they left in full and were removed in transit. The records discriminate between them. Removal in transit would require opening the container — breaking the seal — and would change its weight after the weighbridge reading. The weighbridge agreed with the declared mass, and the seal number runs unbroken through every document to the stripping floor. The evidence therefore points away from removal in transit and toward the first hypothesis: the shortfall existed when the doors were sealed at origin. The investigation turns to the shipper's packing records.

Step 4: The counterfactual. Suppose the opposite record set: no yard weighing, a seal number missing from one interchange record, a handover left unsigned. The same sixty cartons would then sit inside a fog of shared ignorance, with importer, carrier, and shipper each insisting the loss happened elsewhere, and no party able to show where its own custody ended clean.

Step 5: The outcome. With the records intact, the importer raises a short-shipment claim against the seller with the evidence attached, the seller's packing records confirm the short pack, and the resolution proceeds without a year of argument.

What does this example demonstrate? Weight, seals, and signatures are not administrative decoration. They are the instruments an investigation uses to locate where integrity broke. The habit of questioning a weight discrepancy is exactly this: treating a number that disagrees as a question the records must answer.

Key lesson: records written at the time locate the break in the chain; records written after the fact locate nothing.

WORKED EXAMPLE

An exporter prepares for trusted-operator certification

A shea butter exporter in Tamale supplies buyers in Europe. A new buyer, reviewing its suppliers, asks a question the exporter has never been asked: what supply chain security certification does your operation hold? The buyer's customers increasingly want assurance that their suppliers' chains are secure; under a trusted-trader programme of the kind developed internationally in the WCO SAFE Framework, that assurance is demonstrated by certification from a customs administration where such programmes operate. The exporter resolves to prepare. This account is illustrative.

Step 1: Understanding what is assessed. Working with its customs broker, the exporter learns the assessment logic in outline: the administration examines whether the business can demonstrate secure premises, controlled procedures, screened personnel, and records that let it reconstruct its consignments' handling — the sound security controls and compliance record the framework requires.

Step 2: A gap review of the premises. The packing facility is walked against this lesson's physical security measures. The findings: the perimeter fence is sound, but the gate is unlogged; the yard

is lit, but the packing hall door stands open through shifts; finished stock sits unlocked. Remedies follow: a gate register, closed-door discipline, and a locked store with keys issued to two named supervisors.

Step 3: Procedures made visible. Sealing and dispatch become documented steps: every export container's seal number recorded on the packing record and the transport documents; every collection verified against the driver's identity and the collection authorisation; every handover signed. A simple seal register joins the dispatch file.

Step 4: Personnel and access. Staff in sensitive roles — packing, sealing, documents — are identified for screening; badges are issued; a visitor register with escorted access begins immediately.

Step 5: Records assembled as evidence. The handover-and-seal record already existed in fragments; consolidation turns it into the demonstration material. The application is not a story about being secure; it is a trail showing it.

In the illustrative outcome, the assessment examines the documentation, visits the premises, watches a dispatch against the written procedure, and reviews the compliance record. The certified operator gains facilitation — faster processing, fewer inspections — and, through mutual recognition between customs administrations, recognisability on further trade lanes. And the buyer's question is now answered with a certificate rather than an apology.

What does this example demonstrate? Trusted-operator status is not bought; it is the documented practice of this lesson's disciplines, assessed. The exporter's preparation added little that a secure operation should not already have had. The change was making each control visible, consistent, and recorded.

Key lesson: certification is integrity discipline made examinable — the same habits, written down and consistently kept.

IMPORTANT NOTE

Security is a chain: your partners' controls are your controls

Nothing in this lesson protects a consignment that spends most of its journey in someone else's hands unless those hands are trustworthy. Security is a chain: a sealed, supervised, perfectly recorded dispatch can be undone by an unvetted haulier, an unlocked subcontractor warehouse, or a destination agent that signs for containers without looking at seals. In security terms, an operator is the weakest partner it employs.

Two practical consequences follow.

First, partners are chosen, not inherited. Security belongs on the checklist when carriers, warehouses, and agents are selected: what access control operates at their facilities; how drivers are verified and identified; how handovers and seals are recorded; whether their records

survive long enough to be useful in an investigation. The chain-of-custody block's known-parties rule is this checklist in motion, and Module 15, on logistics strategy and service-provider selection, returns to how such criteria are weighted and contracted; this lesson supplies the security questions to ask.

Second, partners are audited, not assumed. A partner secure at selection can drift; periodic review of partners' facilities and records keeps the chain's weakest link from silently thinning. Trusted-trader certification helps here too: a current certification means a third party more demanding than most commercial audits has examined the operator's controls.

The flip side is equally true: being a strong link raises an operator's value to every chain it joins. Businesses route cargo toward partners whose records they can trust, and trusted-operator status is increasingly how that trust is signalled.

IMPORTANT NOTE

A seal may be broken, but never silently

The seal discipline says a broken or mismatched seal is a stop event — but some stops are legitimate. A customs examination may require a container to be opened in transit; an emergency may require access to the load. The discipline does not forbid breaking a seal. It forbids breaking one without a story.

The rule: whenever a seal must be broken and replaced before delivery, the event is documented at the time — who broke it, on whose authority, why, when and where, who witnessed it, and the numbers of both the old seal and the new. The replacement seal's number goes onto the documents that travel with the consignment, so that every later checker sees continuity, not contradiction. A customs inspection leaves its own official record; where the break is a commercial one, the shipper or agent creates the equivalent.

Why such weight on paperwork? Because the next person to check that seal is a stranger to your emergency. They see a seal whose number appears nowhere in the documents, and they must treat it as a stop event: photograph it, report it, hold the goods. Your undocumented legitimate replacement is indistinguishable from tampering, and it costs everyone downstream.

The principle extends beyond seals. Any intervention that changes the physical or documentary state of a consignment — opening, repacking, resealing, correcting a document — is legitimate only when it is witnessed and written down. A seal story that cannot be told is an integrity failure, and the story that must be tellable is yours.

IMPORTANT NOTE

The innocent shipper answers for what is found in the container

The container carries your name and your declaration. If prohibited or undeclared goods are found inside it — inserted after packing by someone else in the chain — the discovery happens against you first. To customs, to the carrier, and to your buyer, you are the party that declared

the load. Innocence is an outcome you must be able to demonstrate, not a status anyone grants on request.

This is why the declaration disciplines of the air cargo acceptance lesson (Lesson 7.2) and the dangerous goods lesson (Lesson 11.2) are security disciplines as much as compliance ones. Knowing exactly what you packed — every package on the list, and nothing on the floor that is not on the paper — is the foundation on which every other control in this lesson stands. A shipper that cannot account for its own load cannot show that an insertion happened to it rather than because of it.

The protective sequence is short: supervise packing, so you know what entered the container; verify the pack against the list before sealing, so the record is complete when you know most; seal promptly with a recorded high-security seal, so that moment is frozen into evidence; hand over only to known parties, so the frozen evidence is protected; and treat every anomaly — a seal that puzzles you, a weight that disagrees with the declared verified gross mass, a package that is not on the list — as something to question and report, never something to explain away.

Integrity discipline is self-protection: it is how the honest operator proves it was honest when someone else was not.

Practice questions (Lesson Checks)

Practice Question 1

At delivery to a warehouse in Accra, a driver compares the container's seal number against the bill of lading before opening, as company procedure requires, and finds that the two numbers differ. The seal itself appears intact and undamaged.

Which action best reflects the seal discipline this lesson teaches?

1. Cut off the mismatched seal, fit a new seal, record the new number on the delivery documents, and complete the delivery so the vehicle is not delayed.

2. Accept the delivery, because the seal is intact and the consignment documents were correct when the container left the load port.

3. Stop before opening the container, photograph the seal and the document, record the discrepancy, and report it to the agent so the carrier and, where appropriate, the authorities can investigate while the evidence is preserved. ✓

4. Refuse the delivery, return the container to the port unopened, and notify the insurer so that a claim can be filed for the cargo's full value.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

A seal number that does not match the documents means the container's history since the last good record cannot be vouched for. Under this lesson's discipline that is a stop event: the consignment is not accepted and stripped while its story is unknown. The driver preserves the position exactly as found — seal untouched, container unopened — documents it with photographs and a written record, and reports promptly so that the parties holding the gate records, interchange records, and investigative authority can reconstruct what happened. This is how the first worked example's discovery stayed attributable instead of becoming an unexplained shortfall.

Answer 1 is the silent break-and-replace the lesson condemns: it destroys the physical evidence and fabricates documentary continuity. Answer 2 accepts a container whose integrity cannot be attested on the strength of appearances alone. Answer 4 overreaches: returning the container without documenting anything, and notifying the insurer before the carrier, skips the evidence and reporting steps on which both the investigation and any later claim depend.

The takeaway: when the seal and the documents disagree, the consignment waits and the evidence speaks first.

Practice Question 2

An importer strips an import container and finds 60 cartons missing against the packing list. The records show that the terminal weighbridge at the load port agreed with the shipper's declared verified gross mass, that the seal number runs unchanged through the bill of lading, the equipment interchange receipt, and the delivery check, and that every handover was signed.

Which statement best describes what this record set shows?

1. The shortage must have occurred during the ocean carriage, so the carrier is liable by default for the missing cartons.

2. The weighbridge and seal records are administrative formalities that shed no light on where the shortage arose.

3. Because the container left the load port at its declared weight with an unbroken seal sequence, the evidence points away from removal in transit and toward a shortfall that already existed when the container was sealed at origin, directing the investigation to the shipper's packing. ✓

4. Because the delivery driver held the vehicle when the shortage was found, the driver is the most likely cause and should be suspended pending the outcome.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

This is the reasoning of the second worked example. Removing goods from a sealed container in transit requires opening it, which would break the seal and change the container's weight; here the weighbridge agreed with the declared verified gross mass and the seal number is continuous through every document to the stripping floor. Transit removal has no room in this record. The shortfall must therefore predate the sealing — a short pack at origin — and the investigation moves to the shipper's packing records, where, in the worked example, it found confirmation.

Answer 1 assigns liability by assumption rather than by evidence; the records point away from the carriage leg entirely, and no party is liable by default. Answer 2 dismisses precisely the instruments — weight and seal continuity — that let an investigation locate the break in the chain. Answer 4 selects a suspect by proximity instead of by record; the signed handovers and matching seal tend to exclude the delivery leg rather than incriminate it.

The takeaway: records written at the time locate where integrity broke; assumptions about blame locate nothing.

Practice Question 3

A freight forwarder in Kumasi is explaining trusted-trader certification to a client whose overseas buyer has asked about supply chain security.

Which statement best describes what certification under the trusted-trader concept means?

1. It exempts a certified company from all customs formalities and physical inspections in every country it trades with.

2. It is a paid fast-lane arrangement that any company can obtain by paying the applicable fee, whatever its security controls.

3. It is a certification by the customs administration that an operator's security controls and compliance record are sound, earning facilitation such as faster processing and fewer inspections, with mutual recognition arrangements allowing the benefits to be recognised between customs administrations. ✓

4. It is an insurance arrangement under which the certifying body compensates the operator for losses caused by cargo theft or tampering.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

Under the WCO SAFE Framework of Standards, the trusted-trader concept — known internationally as Authorised Economic Operator status — is a partnership. The operator demonstrates secure premises, controlled procedures, screened personnel, and reliable records, and its customs administration certifies the result, returning facilitation in the form of faster processing and fewer inspections. Mutual recognition arrangements extend the certification's usefulness across borders, which is why the client's overseas buyer can rely on it. The third worked example showed that what is examined is exactly the documented practice of this lesson's disciplines.

Answer 1 overstates the benefit: facilitation reduces friction, but it abolishes nothing, and certified operators remain subject to controls. Answer 2 inverts the logic — certification is earned through demonstrated controls and a compliance record, not purchased, because payment proves nothing about security. Answer 4 confuses certification with insurance: the status involves no compensation and no cover, and protection against loss remains a matter for the cargo insurance arrangements the previous lesson described.

The takeaway: trusted-trader status converts documented integrity discipline into faster, easier movement through international trade.

Fact-check appendix (audit notes — not for entry)

Definitions of supply chain security, cargo integrity, cargo theft, cargo tampering, container seal, high-security seal, access control, chain of custody, shipment traceability, security inspection, cargo diversion, audit trail, and security incident — source: curriculum lesson brief module-11/lesson-11.3.md (key terms, taught as defined)

The WCO SAFE Framework of Standards was adopted in June 2005; under its customs-to-business pillar an operator with a sound security/compliance record is certified by customs and receives facilitation benefits, and mutual recognition arrangements (MRAs) extend benefits across borders — source: research sheet trade-transport-standards.md section 12 (WCO, https://www.wcoomd.org/-/media/wco/public/global/pdf/topics/facilitation/instruments-and-tools/tools/safe-package/safe-framework-2025_en.pdf); taught generically, no country programme names

Security is a recognised category in the logistics risk taxonomy (supply-side, transport, customs/regulatory, financial, security, natural), and the risk-management cycle runs identify, assess, mitigate/treat, monitor and review — source: research sheet logistics-scm-frameworks.md section 11

Attacks on shipping in the Red Sea from late 2023 led major container lines to reroute widely around the Cape of Good Hope — source: research sheet logistics-scm-frameworks.md section 11 (stated qualitatively; no magnitudes, loss figures, or theft/piracy statistics anywhere in the lesson)

Three-way match of purchase order, goods receipt, and invoice before payment — source: research sheet logistics-scm-frameworks.md section 3

Document checks, three-way match, and approved carrier/broker lists as controls against trade-document manipulation and fraud — source: research sheet logistics-scm-frameworks.md section 13 (qualitative; no specific allegations)

Track-and-trace via scan/status events — source: research sheet logistics-scm-frameworks.md section 9 (qualitative)

Verified gross mass: the shipper must obtain and communicate the verified gross mass of every packed container under SOLAS VI/2, and no container without a VGM may be loaded — source: research sheet trade-transport-standards.md section 7 (IMO, https://www.imo.org/en/OurWork/Safety/Pages/Verification-of-the-gross-mass.aspx); cross-referenced to the VGM key term of lesson brief module-07/lesson-7.3.md

Customs risk management routes declarations toward release without inspection, document check, or physical examination; the channel model is used in many administrations including Ghana's ICUMS — source: research sheet trade-transport-standards.md section 12

The Customs Division of the Ghana Revenue Authority has a three-fold responsibility: revenue mobilisation, facilitation of legitimate trade, and public security and safety — source: research sheet ghana-trade-customs.md section 2

ICUMS assigns every consignment a Unique Consignment Reference (UCR) supporting end-to-end tracking — source: research sheet ghana-trade-customs.md section 1

Customs transit regimes move goods under customs seals with a transit document and guarantee system; the ECOWAS Inter-State Road Transit (ISRT) arrangement is the West African analogue of TIR, and Ghana applies ISRT for transit cargo — source: research sheet trade-transport-standards.md section 12 (taught generically)

The equipment interchange receipt acknowledges the transfer of a container and records relevant condition details — source: lesson brief module-09/lesson-9.3.md (EIR key term); seal numbers carried on the bill of lading and interchange documents — source: general industry practice, consistent with trade-transport-standards.md section 4

Call-back verification of changed instructions, approved-party lists, personnel screening and access withdrawal on departure, visitor registers, gate logs, supervised loading, and prompt incident notification supporting claims — source: general business knowledge

All worked-example quantities, weights, seal numbers, and outcomes are illustrative for teaching purposes per writing spec rule 5.1; no statistics on cargo theft or piracy, and no named national security programmes, are used anywhere in the lesson

Cross-references: Lesson 7.2 Air Cargo Operations and Cargo Acceptance; Lesson 7.3 Containerisation: FCL, LCL, and Container Types; Lesson 9.3 Cargo Release, Delivery Orders, and Terminal Procedures; Lesson 11.1 Cargo Insurance and Risk Allocation; Lesson 11.2 Dangerous Goods and Special Cargo Handling; Lesson 11.4 Managing Shipment Delays, Damage, and Discrepancies; Module 8 warehouse record-keeping discipline; Module 15 service-provider selection — source: curriculum lesson briefs

Lesson 11.4 — Managing Shipment Delays, Damage, and Discrepancies

Summary

This lesson examines the procedures used to identify, investigate, and resolve operational problems during cargo transportation. Learners will explore shipment delays, damaged goods, missing cargo, quantity differences, and documentation discrepancies. The lesson emphasises incident reporting, evidence collection, stakeholder communication, corrective action, and the importance of maintaining accurate records when managing logistics exceptions.

Learning objectives

- Identify common causes of shipment delays, cargo damage, and discrepancies.

- Explain how operational disruptions affect logistics performance and customer service.
- Describe procedures for identifying and reporting shipment exceptions.
- Explain the importance of documenting cargo condition and discrepancies.
- Identify the evidence required to investigate cargo loss or damage.
- Describe effective communication practices during shipment disruptions.
- Distinguish between corrective and preventive actions.
- Explain the role of carriers, freight forwarders, insurers, and customers in resolving shipment problems.
- Develop a basic response plan for a delayed or damaged shipment.

Key terms

Shipment Exception — An event or condition causing a shipment to deviate from its expected transportation or delivery process.

Shipment Delay — A situation in which cargo movement or delivery occurs later than the planned or agreed schedule.

Cargo Damage — Physical deterioration or impairment affecting goods during handling, transportation, or storage.

Short Shipment — A shipment containing fewer goods than the quantity specified in the relevant shipping or commercial records.

Over Shipment — A shipment containing more goods than the quantity specified in the relevant shipping or commercial records.

Documentation Discrepancy — An inconsistency or error identified within or between shipment documents.

Incident Report — A record describing an unexpected event, its circumstances, and its observed consequences.

Exception Management — The identification, investigation, communication, and resolution of disruptions affecting normal operations.

Root Cause Analysis — A structured process for identifying the underlying causes of a problem.

Corrective Action — A measure taken to eliminate the cause of an identified problem or nonconformity.

Preventive Action — A measure taken to reduce the likelihood of a potential problem occurring.

Cargo Survey — An examination of goods to assess their condition, quantity, damage, or other specified characteristics.

Claim Notification — Formal communication informing a responsible party or insurer of a potential loss or damage claim.

Contingency Plan — A prepared course of action for responding to disruptions or unexpected operational events.

Teaching content

KEY CONCEPT

Exceptions are normal; exception management is the discipline

However carefully a shipment is planned, it is executed across weeks of transit, changing weather, congested infrastructure, dozens of handovers, and the paperwork of several organisations. Somewhere in that span, something will deviate from plan. A shipment exception is an event or condition causing a shipment to deviate from its expected transportation or delivery process — a vessel that sails late, cartons that arrive crushed, a container that strips four cartons short, a packing list that disagrees with the invoice. The definition's breadth is the point: an exception is any departure from the expected process, whether or not anyone is at fault and whether or not money is lost.

Exception management is the identification, investigation, communication, and resolution of disruptions affecting normal operations. The four verbs carry the discipline. Identify: know about the deviation early, from tracking and status events rather than from an angry telephone call — Lesson 5.4, Shipment Tracking and Customer Communication, built that visibility. Investigate: establish what happened from records and evidence, not from assumption. Communicate: tell the parties who must know, early and factually. Resolve: fix the shipment's immediate problem, then fix the cause so the exception does not return.

Two convictions underlie the whole lesson. First, exceptions are a normal operating workload, not a disgrace: every logistics operation has them, and mature operations are distinguished not by having none but by how they respond. Second, an unmanaged exception is a service failure by definition. Performance measurement makes this concrete: on-time-in-full performance counts a delivery only when it arrives both on time and complete, so a consignment that is late, short, or damaged fails the measure whatever the reason — weather, another party's error, or plain bad luck. The customer experiences the exception, not its explanation. What the operation controls is the quality of its response, and that response is a method, not a mood.

KEY CONCEPT

The incident response framework: six movements

Module 11 opened with risk managed as a loop: identify what could go wrong, assess likelihood and impact, mitigate what matters, monitor and review. This lesson extends that loop to the moment a risk has materialised — the incident has happened, and the cycle becomes a response. The response framework has six movements, and every worked example in this lesson walks through them.

1. Detect early. Most exceptions are first visible as a data point: a vessel off schedule, a status event that stops updating, an estimated delivery date that slips. Detection depends on the tracking disciplines of Lesson 5.4. The value of early detection is options: a delay caught at sea can be re-planned; the same delay discovered on the customer's dock can only be apologised for.

2. Contain and make safe. Safety precedes commerce, always. A leaking drum, a shifted load, a damaged package of dangerous goods — the dangerous goods lesson (Lesson 11.2) governs here — changes the order of every following step: people are protected and the situation is stabilised before any claim is discussed.

3. Preserve evidence. Photographs, seals, packaging, documents, remarks on delivery papers — taken before anything is moved, cleaned, repaired, or repacked. Evidence preserved at discovery is the raw material of every later step; evidence tidied afterwards is worth nothing.

4. Notify the right parties. Customer, insurer, carrier, agent, and authorities as applicable — each within its own process. Who must know, and what to say, is treated later in this lesson.

5. Resolve and record. Decide the operational fix — reroute, replace, expedite, recover — execute it, and write the incident report while memory is fresh.

6. Learn and improve. Establish the root cause, take corrective and preventive action, and feed the finding back into the risk cycle. Every incident teaches the system something; the framework exists so the lesson is captured rather than absorbed as loss.

KEY CONCEPT

Common causes of shipment delay

A shipment delay is a situation in which cargo movement or delivery occurs later than the planned or agreed schedule. The value of a cause list is diagnostic: when a delay appears, it tells an operator where to look first and what to tell the customer. The recurring causes are these.

1. Weather and natural events. Storms close ports, fog grounds flights, flooded roads stop trucks — nobody's fault, but the consequences still need managing.

2. Congestion. Ports, terminals, and border posts work to finite capacity; when demand exceeds it, queues form and vessels wait at anchorage. From early 2020 through roughly 2021–2022, congestion around the COVID-19 pandemic disrupted schedules worldwide as demand shifted and containers accumulated in the wrong places.

3. Transhipment misses. A container that misses its connecting vessel or flight rolls to the next departure; one missed connection can add days or weeks to a journey planned around a tight schedule.

4. Documentation problems. A consignment whose documents are incomplete, inconsistent, or late can be held at origin before it sails or at destination before it is released — the document disciplines of Module 6 exist to prevent precisely this cause.

5. Customs holds. Examinations, permit queries, and valuation questions stop cargo administratively. The risk-management and clearance lessons of earlier modules explain why selection for examination is normal and how preparation keeps it short.

6. Equipment and capacity shortages. Containers in the wrong region, vessel space overbooked, too few trucks at the port — capacity problems upstream become delays downstream.

Each cause becomes, further downstream, the same customer experience: a promise that slips. Three disruptions deserve to be remembered as proof that delay is systemic rather than personal failure. The Ever Given ran aground in the Suez Canal on 23 March 2021 and was freed on 29 March 2021, blocking the waterway for six days and seven hours while hundreds of vessels queued. From December 2023, attacks on shipping in the Red Sea led major container lines to reroute widely around the Cape of Good Hope, substantially lengthening voyages between Asia and Europe. No trader caused those events; the delays arrived anyway. That is why the professional response is a method, not an apology — and the method comes next.

KEY CONCEPT

Managing a delay: re-plan the promise, mitigate, record

When a delay is detected, the management pattern has three moves, in this order.

1. Re-plan the customer promise early. The customer's promise — the delivery date they built their own plans around — is now at risk, and the worst outcome is for them to discover it themselves. The re-planned promise has three parts: a revised estimated time of arrival, the cause stated plainly, and the options the customer can choose among. This is the communication discipline of Lesson 5.4 applied under pressure: news given once, early, with facts. A revised promise the customer can plan around is worth more than an original promise quietly missed.

2. Mitigate downstream. The question is what the delay does to everything after it, and what can be done about that now. The options include drawing on alternative stock — safety stock, the buffer held against variability in supply and lead time, exists partly for exactly this moment, and the service level the operation commits to, as inventory control teaches (Lesson 8.3), prices how often that buffer will be needed; supplying a critical portion early by a faster mode while the balance follows; rerouting the remaining legs; or renegotiating delivery windows. Mitigation spends money to protect service, so it is chosen with total-cost eyes: the inventory carrying-cost logic of Lesson 10.1 reminds us that weeks of additional transit mean capital afloat longer, even before any extra freight is paid.

3. Record. The delay closes with a record: cause, duration, decisions taken, costs incurred, communications sent. The record is what turns one delay into the next contingency plan.

A contingency plan is a prepared course of action for responding to disruptions or unexpected operational events. It pre-decides the first hours: who is notified, by whom, from which options, in which order. The difference between an operation that improvises and one that executes a contingency plan is most visible on the first day of a disruption — and every later move is easier when the first was rehearsed on paper.

KEY CONCEPT

Cargo damage and the three discovery moments

Cargo damage is physical deterioration or impairment affecting goods during handling, transportation, or storage. Its common causes are mechanical and environmental: rough handling, crushing under badly planned stowage, dropped units, moisture and water entry, temperature extremes, contamination, and vibration over long road legs. Prevention belongs to the packaging, stowage, and handling choices made long before the journey — and reviewed after every incident, as the lesson's final movement requires. This block is about the moment damage is discovered, because the discovery moment decides the evidence.

Damage is discovered in three moments.

1. At delivery, externally visible. This is the moment of maximum leverage. The receiving discipline of Lesson 8.2 requires inspection before acceptance, and its most consequential habit is this: damage seen before signing is noted on the delivery document before the signature is given. A clean receipt attests that goods arrived in apparent good order; a written remark — cartons crushed, packaging torn, units stained — converts the document from the carrier's evidence into the consignee's. Module 6 introduced clean and claused transport documents on the carrier's side; the same logic operates at your own gate. A signature given without remark is a statement that all was in order.

2. At unloading or stripping. Concealed damage emerges as packaging is opened: a carton intact outside, its contents broken within. The discipline is to stop at the discovery, photograph the package exactly as found, keep the packaging, and record the finding before any further handling.

3. Later, after acceptance. Goods accepted clean and found damaged when put to use — the hardest case, because the consignee must then show the damage existed on arrival with no contemporaneous remark to help. Report at once to carrier and insurer, preserve the goods and packaging untouched, and expect the insurer to require a survey. The strongest argument for a thorough check at receipt is that it makes the third moment rare.

In every moment: segregate damaged goods from saleable stock, and do not discard anything — the goods and their packaging are evidence.

KEY CONCEPT

Evidence discipline and the cargo survey

An investigation into loss or damage answers four questions: what was shipped, in what condition, who held it along the way, and when did the condition change. The evidence that answers them is assembled from sources the course has already built, gathered into one file.

1. The document set. The transport document, commercial invoice, packing list, and insurance policy or certificate — the set whose cross-checking Lesson 6.4, Document Verification and Common Discrepancies, trained. These prove what was supposed to move, in what quantity, under whose carriage and cover.

2. Condition records at handovers. The equipment interchange receipt of Lesson 9.3 recording container condition and seal status at each interchange; seal numbers running unchanged through the documents; the delivery document bearing the damage remark. Together they reconstruct the chain of custody and locate the last moment the goods were provably sound — the reconstruction logic of Lesson 11.3 applied to a damage file.

3. Photographs. Wide shots establishing position and context, close-ups showing the damage, labels and seal numbers legible, taken before anything is moved. Photographs are testimony of the moment; notes written afterwards are only a story about it.

4. Physical evidence. The damaged goods and their packaging, retained and segregated — not discarded, not repaired, not repacked. Packaging often carries the cause: crush points, water stains, handling marks.

5. The incident report. A record describing an unexpected event, its circumstances, and its observed consequences — written while memory is fresh, stating facts and leaving conclusions until the evidence is examined.

6. The cargo survey. An examination of goods to assess their condition, quantity, damage, or other specified characteristics. For substantial claims the insurer will require a survey by its surveyor, whose report anchors what happened and how extensive the loss is — the claim sequence of Lesson 11.1 anticipated this. The surveyor surveys what is preserved: a survey of repacked goods finds little and proves less.

The discipline binding all six: preserve first, conclude later. The evidence file is built at the speed of discovery, not at the speed of certainty.

KEY CONCEPT

Two channels for loss: the carrier claim and the insurance claim

When goods are lost or damaged in transit, two claim channels may be open, separately or together. Lesson 11.1, Cargo Insurance and Risk Allocation, established the principle that carrier liability is not cargo insurance; this lesson treats them as the two procedural channels they are.

1. The carrier claim. A claim against the carrier — the shipping line, airline, or road haulier, pursued directly or through the freight forwarder who arranged the carriage — rests on the contract of carriage: the carrier received the goods and issued a transport document describing them, and goods delivered damaged or short are evidence against that receipt. Its supporting evidence is exactly the file of the previous block, with the delivery remark at its centre. Carrier liability is bounded by the carriage contract and the liability rules governing the mode, and those rules attach their own requirements as to when notice must be given.

2. The insurance claim. A claim under the cargo policy, opened by claim notification — formal communication informing a responsible party or insurer of a potential loss or damage claim. The insurer responds within the policy's terms, cover level, and deductible, usually requiring the survey already described; and having compensated, an insurer commonly takes over the claim against the party responsible, so the carrier file still matters.

The channels are distinct but not alternatives. The carrier claim preserves rights against the party that handled the goods; the insurance claim recovers the loss within the policy regardless of how the liability argument runs. A professional file serves both.

Each party around the file has a defined role. The carrier answers for the carriage it performed. The freight forwarder orchestrates: bookings, documents, notices, and the coordination of claims across channels — the forwarder's representative role from Lesson 5.1 applied to adversity. The insurer assesses and compensates within the policy. The customer must be kept informed, because their commercial decisions — replacement orders, sales commitments — hang on the outcome. And the cargo owner owns the whole: no party manages the response unless engaged to do so.

KEY CONCEPT

Discrepancies: quantity, identity, and documents

A discrepancy is any disagreement between what the records say and what the goods or the other records show. Three families cover the ground.

1. Quantity discrepancies. A short shipment is a shipment containing fewer goods than the quantity specified in the relevant shipping or commercial records — ninety-six cartons stripped against one hundred on the packing list. An over shipment is a shipment containing more goods than those records specify — and it is not a windfall: undeclared extra goods create record and customs discrepancies that must be declared and regularised, not quietly absorbed. Quantity discrepancies arise from short packing at origin, mis-counts at dispatch or receipt, unrecorded movements, and theft; a well-run stock reconciliation assumes none of these and investigates before it adjusts, as Lesson 8.4, Warehouse Documentation and Stock Reconciliation, teaches.

2. Identity discrepancies. Wrong goods, the right goods in the wrong variant or specification, cartons belonging to another order mixed into the load. Their causes are usually picking and labelling failures at dispatch; their signature is goods that do not match their own paperwork.

3. Documentation discrepancies. A documentation discrepancy is an inconsistency or error identified within or between shipment documents — quantities that disagree between invoice and packing list, a description that does not match the goods, references that differ between documents. Lesson 6.4's verification discipline — check the set against itself and against the goods before acting — is the defence; the three-way match of purchasing practice (purchase order, goods receipt, and supplier invoice agreeing before payment) is the commercial cousin of the same reflex.

The response to every family begins the same way: treat the discrepancy as a question the records must answer, not a verdict to be absorbed. The reconciliation reflex — records before adjustments — means no stock figure is corrected, no write-off posted, and no claim issued until the discrepancy has been traced through the records. An adjustment made in haste converts a recoverable question into an unrecoverable one. Where internal tracing exhausts the records, escalation follows the chain: the supplier for short packing, the forwarder and carrier for transit losses, each approached with the evidence file attached.

KEY CONCEPT

Communication during a disruption: who must know, and what to say

A disruption can be handled well operationally and still badly relationally, and the difference is communication. The discipline has three parts: the right list, the right content, the right sequence.

The list — who must know now.

1. The customer, first with the operational news: the revised promise with its cause and options, as the delay block set out. Never let a customer learn of their own delay from their own dock.

2. The insurer, by claim notification while the evidence is fresh, whenever loss or damage may mature into a claim.

3. The carrier and the agent, whose own processes — surveys, exception reports, terminal records, recovery of damaged units — start only when told. They also hold records the investigation will need.

4. Authorities as applicable: accidents involving injury, dangerous-goods incidents (Lesson 11.2), and customs matters each bring the relevant authority onto the list, and safety-related notification precedes commercial notification.

Internal escalation sits at the top of the sequence: the incident goes up the management line at discovery, so that decisions about money, customers, and safety are taken at the level that owns them.

The content: facts and consequence, in one consistent version. State what is known, what is not yet known, what is being done, and what the recipient must decide or provide. Avoid

speculation about cause before the evidence supports it — a cause announced early and withdrawn later costs more credibility than no cause at all — and never conceal or soften an exception in the hope that it resolves itself. The version given to the customer, the insurer, and the carrier must be the same version; audiences compare notes, and a discrepancy between accounts becomes a second incident.

The sequence: early beats complete. A short factual message on the day of discovery — what happened, what is still unknown, when the next update will come — outperforms a polished account a week later. Bad news early is a service; bad news late is a betrayal.

KEY CONCEPT

Root cause analysis, corrective action, and preventive action

Resolution is not the end of an incident. The final movement of the response framework asks what the incident taught, and three defined ideas give the question its structure.

Root cause analysis is a structured process for identifying the underlying causes of a problem. The word underlying does the work: the crushed cartons were caused by a forklift's handling, but the underlying cause may be a stacking specification the cartons could not carry, a loading routine that rewarded speed over care, or packaging chosen on price alone. Root cause analysis keeps asking why until it reaches a cause the operation can change, and stops there — not at a person's name. An investigation that ends at blame produces a scapegoat; one that ends at a process produces a change.

Corrective action is a measure taken to eliminate the cause of an identified problem or nonconformity. It looks back at what happened and removes its cause, so that this incident does not recur: revising the stacking specification and enforcing it at loading; adding the missing verification step to the procedure; retraining against the changed procedure.

Preventive action is a measure taken to reduce the likelihood of a potential problem occurring. It looks forward at what has not yet gone wrong and reduces its likelihood: extending the packaging review to the other product lanes that use the same carton before they fail too; applying the revised tally discipline to every receiving point, not only the one that erred.

The distinction is timing and target: corrective action answers an incident that occurred; preventive action answers one that has not — yet. Replacing the damaged goods alone is neither: that is remediation of this consignment, and an operation that stops there has treated the symptom and kept the cause. Mature operations insist that both actions are recorded, each with an owner and a review date, before an incident file is closed.

WORKED EXAMPLE

A damage claim done well: crushed cartons at an Accra warehouse

Adwenpa Distribution Ltd, a distributor of packaged consumer goods in Accra, imports a consignment of 800 cartons of bottled cooking products, discharged at the Port of Tema and

delivered by a hired haulier to the company's warehouse. All quantities and outcomes in this example are illustrative for teaching purposes.

Step 1: Discovery at the gate. The receiving clerk walks the vehicle before signing, as the receiving discipline of Lesson 8.2 requires. Thirty cartons on the top tiers are crushed; eight of those are stained and sticky. The driver waits with the delivery note.

Step 2: The note before the signature. The clerk writes on the delivery document: "30 cartons crushed, 8 stained — noted on receipt at 10:40," and only then signs; the driver countersigns the remark. The document is now evidence in the consignee's hands. Had the clerk signed clean and complained afterwards, the consignment would carry the burden of proving the damage happened before delivery.

Step 3: Preservation. Photographs: the vehicle with its doors open, the pallet positions, wide and close shots of the crushed cartons, labels legible. The damaged cartons and their packaging are unloaded and segregated in the marked damaged-goods area — retained, unopened where intact, nothing discarded.

Step 4: Notification. The same day: the forwarder who managed the import is informed and asked to notify the ocean carrier; claim notification goes to the insurer under the distributor's cargo policy; an incident report is opened recording what was found, when, by whom, and in what condition. The haulier is given a copy of the noted document.

Step 5: The survey. The insurer arranges a cargo survey. The surveyor examines the segregated cartons with their packaging, reads the photographs and the noted delivery document, and reports: crushing consistent with inadequate stowage — cartons stacked beyond their strength beneath heavier units.

Step 6: Settlement and the loop. The insurance claim settles within the policy terms; the noted file supports the parallel carrier claim pursued through the forwarder. Root cause analysis points past the forklift to the stowage specification. Corrective action: a written stowage and stacking standard agreed with the supplier — heavy units low, within the cartons' strength. Preventive action: the same standard extended to the distributor's other supplier lanes before they fail too. The incident file closes with both actions recorded and owned.

What does this example demonstrate? Every movement of the response framework, executed in order: detection at the gate, containment, evidence preserved, the right parties notified, survey and resolution — and the loop closed with a cause and two changes.

Key lesson: the remark written before the signature is the cheapest evidence an importer will ever own.

WORKED EXAMPLE

A systemic delay handled: a Singapore desk, December 2023

A freight forwarder in Singapore manages ocean consignments from Asia to Europe for garment and electronics customers. From December 2023, following attacks on shipping in the Red Sea, major container lines begin widely rerouting vessels around the Cape of Good Hope, substantially lengthening voyages between Asia and Europe. This account is illustrative; the disruption and its dates are matters of record.

Step 1: Detection. Carrier advisories arrive: sailings on the affected routes are being rerouted. The operations desk pulls every booking on the affected services — for illustration, forty consignments in transit or awaiting loading — and marks each customer's commitment date against the longer voyage times.

Step 2: Re-plan the promises. No customer learns of this from a missed delivery. Each affected customer receives the same three-part message: the cause stated plainly (a named, dated disruption to the shipping routes), a revised estimated arrival built on the rerouted schedules, and the options — hold and accept the longer transit, supply from alternative stock where they hold it, or expedite a critical portion by air where the business case survives.

Step 3: Mitigation and re-pricing. One customer, a garment buyer, holds several weeks of safety stock and rides the delay at no extra cost — the buffer bought for exactly this. Another, with components feeding a production line, expedites a small critical portion by air and lets the balance sail the long way round. The forwarder walks each customer through the in-transit inventory consequence in the carrying-cost language of Lesson 10.1: weeks more at sea mean capital afloat longer and replenishment slower — a real cost even where the freight charge is unchanged. Quotations and transit-time commitments are re-based.

Step 4: Record and adapt. The event and every decision are logged. The lasting change is structural: sailing-time assumptions in quotations and customer commitments are updated permanently, not treated as a one-off, and the contingency plan gains a first-day checklist — identify affected bookings, revise each promise with cause and options, re-price the inventory impact — so that the next route closure finds a procedure, not a scramble.

What does this example demonstrate? A systemic delay — nobody at this desk caused the rerouting, and no apology was owed for it — is managed through the same framework as a single damaged pallet: detect early, re-plan the promise with cause and options, mitigate with an eye on total cost, record, and change the assumptions permanently.

Key lesson: when the system shifts, the promise must be re-planned before it is missed — and the assumptions behind it must never silently return to the old map.

WORKED EXAMPLE

Ninety-six against one hundred: a shortage investigated in Kumasi

A hardware distributor in Kumasi receives a twenty-foot container of floor tiles, discharged at Tema and hauled inland. During stripping at the distributor's warehouse, the count arrives at 96 cartons against 100 on the packing list. All figures in this example are illustrative for teaching purposes.

Step 1: Stop before adjusting. The warehouse supervisor records 96 on the strip sheet and posts nothing: no write-off, no stock adjustment, no telephone demand. Records before adjustments — the reconciliation discipline of Lesson 8.4. A claim issued now and withdrawn next week teaches the carrier's claims desk to take this company's claims lightly.

Step 2: Reconstruct the chain. The file answers first for the transit. The container's seal was intact and its number matched the documents at gate-out, recorded on the equipment interchange receipt in the manner of Lesson 9.3, and again at arrival in the seal discipline of Lesson 11.3; every handover was signed. Goods removed in transit would have required breaking that seal, and the record shows no such event. The transport chain is cleared by evidence, and the question narrows to packing at origin or counting at strip.

Step 3: Recount inside the warehouse. Before writing abroad, the supervisor re-examines the container itself. The strip crew had counted at the doors only. A full tally — the final pallet row lifted, the nose of the container swept and inspected — finds four cartons standing behind it, invisible from the doorway. The physical count is 100.

Step 4: Resolve and record. The strip sheet is completed at the true count, the discrepancy note is closed with the finding, and the stock record now reflects a count that was verified rather than assumed.

Step 5: The loop. Root cause analysis: the counting method observed a single point and treated the doorway's view as the container's contents. Corrective action: the strip procedure is rewritten — a tally recorded per lift on the strip sheet, a second count at container clearance, and the strip sheet signed before any stock figure is posted. Preventive action: the same two-point tally extended to the receiving bay's carton checks, and a written rule that stock records move only on completed count documents. The incident file closes with both actions owned and dated.

What does this example demonstrate? The reconciliation reflex in full: a discrepancy is a question; the records and the chain reconstruction narrow it; the physical recount answers it — and the procedure that allowed the doubt is corrected and generalised before the file closes.

Key lesson: investigate before you adjust — the cheapest claim is the one never made, and the most expensive record is the one written to match a guess.

IMPORTANT NOTE

Never amend evidence to fit a story

When an exception is embarrassing — the damage looks like the warehouse's own fault, the shortage might be our count, the delay was our document error — a temptation appears: tidy the scene, repackage the cartons, correct the document quietly, backdate a note, adjust the stock figure so the paperwork balances. Every one of these destroys the only asset the investigation has. Photographs, documents, seals, and physical packaging tell what happened; an amended record tells only that someone intervened, and invites the question of what else was interfered with.

The rules are short. Photograph before touching; nothing is cleaned, repaired, or repacked before the photographs are taken. Keep the originals: the noted delivery document, the strip sheet as first written, the interchange records. Where a record must be corrected, the correction is a further entry — what was wrong, what is right, who corrected it, when, and why — never an erasure. Damaged goods and their packaging are evidence until the survey or the agreed process releases them; they are not to be discarded, and no driver should be allowed to take the damaged ones away. Stock figures are adjusted only after the investigation, on the strength of it.

The deeper point is credibility. An operator whose records are found amended once will have every record doubted thereafter — by insurers pricing a claim, by carriers resisting one, by auditors and customs after them. A story that needs doctored evidence to support it is a confession written slowly. The evidence, left unamended, usually tells a fairer story than the one fear was about to draft.

IMPORTANT NOTE

Late notification narrows your options

This lesson deliberately teaches no day-counts, because there are none to teach: carriers, forwarders, and insurers each run their claims to their own processes, set in their contracts, their conditions of carriage, and their policy wordings, and the requirements differ from party to party and trade to trade. The structural facts hold everywhere. Every claims process expects notice while the evidence is fresh; every one of them treats late notice as a reason to question what is claimed.

Late notification narrows options in three ways. Evidence decays: packaging is discarded, scenes are cleaned, vehicles are released, and memories blur into goodwill. Rights weaken: a damage claim asserted against a carrier long after a clean signature invites the answer that the goods were sound when custody passed — and the carriage rules attach their own time requirements to notice, as Lesson 11.1 noted. Recovery narrows: an insurer whose policy conditions on prompt notice are not met may reduce or decline the very claim the policy seemed to promise — the policy's conditions are part of what is bought.

The professional rule follows. Notify each channel within its process as soon as the facts are known — not the conclusions, the facts: what was found, when, and in what condition. Where it is genuinely uncertain whether a loss will mature into a claim, notify anyway to preserve the position; a notification that turns out to be unnecessary costs a letter, while a deadline discovered to have passed costs the claim. Where a counterparty's process is unknown, establishing it is the agent's or insurer's first task — today, not at settlement.

IMPORTANT NOTE

Every incident ends with a root cause and a change, or it will repeat

An incident file can close at the wrong moment in three ways. It can close at blame: a driver is dismissed, a supplier is scolded, everyone feels the matter is settled — and nothing about the process changed, so the same failure returns under a new name. It can close at settlement: the claim is paid, the relief is real, and the cause is never examined, so the operation has converted its own incident into money without learning from it. It can close at fatigue: the report half-written, the actions noted, the file drifting. All three close the file; none closes the loop.

The discipline this lesson ends on is a definition of done. An incident is finished when four things exist: the incident report, the established root cause, the corrective action that removes that cause, and the preventive action that narrows the family of similar risks — each with an owner and a review date, and the finding fed back into the risk cycle this module opened with, so that identify, assess, mitigate, and monitor run on updated information.

Read this way, the module is one continuous argument. Lesson 11.1 priced the risks and insured what prevention could not stop. Lesson 11.2 controlled the risks that travel with dangerous and special cargo. Lesson 11.3 secured the chain against interference. This lesson supplied the response for when a risk materialises anyway — and its final movement is what makes the whole module a loop rather than a set of subjects: every incident teaches the system, and an operation that lets its incidents teach is quieter, safer, and cheaper year by year than the one that merely survives them.

Practice questions (Lesson Checks)

Practice Question 1

A distributor in Takoradi receives a part-load of imported glass tiles. While the driver waits with the delivery note, the receiving clerk sees that several cartons are crushed and one is wet. Company procedure is unclear, and the driver insists the goods were fine when he loaded them.

Which action best reflects the discipline this lesson teaches?

1. Sign the delivery note clean so the driver is not delayed, photograph the cartons afterwards, and raise any claim later if the damage turns out to be significant.

2. Note the crushed and wet cartons on the delivery document before signing, photograph the damage as found, retain the damaged goods and packaging for survey, and notify the forwarder and insurer so that both claim channels can open with the evidence intact. ✓

3. Refuse the entire delivery, send every carton back unrecorded, and ask the supplier to send a complete replacement consignment.

4. Accept the delivery, repack the intact tiles into sound cartons, discard the crushed ones, and deduct their value from the supplier's next invoice.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

Delivery is the discovery moment of maximum leverage, and Answer 2 uses it in the lesson's order: the damage remark goes on the delivery document before the signature, because a clean signature attests apparent good order and hands the consignee the burden of proving when the damage occurred. Photographing the goods as found and retaining them with their packaging preserves the evidence a cargo survey needs, and prompt notification to the forwarder and the insurer opens both claim channels — the carrier claim and the insurance claim — on a file that can carry them.

Answer 1 commits the clean-signature error at the worst possible moment: the driver disputes the damage, so without the contemporaneous remark the later photographs carry little weight, and both the carrier's and the insurer's processes will ask why nothing was noted at receipt. Answer 3 abandons evidence and commercial position together: refusal without record leaves the damage undocumented and the relationship damaged, and nothing in the lesson supports rejecting a whole load because part of it is damaged. Answer 4 destroys the evidence by repacking and discarding, and invents a remedy the lesson's procedures do not contain — the unilateral deduction bypasses both claim channels and the supplier's answers.

The takeaway: remark before signature, photograph before touch, notification before the day ends — and evidence that stays exactly as found.

Practice Question 2

A freight forwarder learns on Tuesday that a vessel carrying a client's production-line machinery will arrive significantly later than the promised date, because congestion at a transhipment port caused the container to miss its connection. The client has not yet heard anything.

Which statement best describes the professionally correct response?

1. Wait until the container is safely on another vessel before contacting the client, so that the client receives one complete and final story rather than changing information.

2. Inform the client that the delay is entirely the shipping line's fault and that no revised arrival date can be given until the goods actually arrive.

3. Contact the client early with a revised estimated arrival, the cause stated plainly, and the available options — such as expediting a critical portion by a faster mode or drawing on alternative stock — while recording the event and its decisions in the delay record. ✓

4. Cancel the ocean booking at once and move the whole consignment to air freight to protect the promised date, informing the client after the change has been made.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The delay-management pattern has three moves in order: re-plan the customer promise early — a revised estimated arrival, the cause, and the options — then mitigate downstream with total-cost judgement, then record. The early re-planned promise is what lets the client adjust production plans while choices still exist; the options preserve the client's decision rights; and the record turns this incident into the next contingency plan.

Answer 1 sounds considerate but inverts the communication principle: it chooses completeness over earliness, and its certain result is the client discovering the delay from their own dock, with every option except apology already gone. Answer 2 offers blame and uncertainty where the discipline offers facts and choices — a cause stated plainly is not an accusation, and producing a workable revised date from the rerouted schedule is precisely the forwarder's job. Answer 4 makes a major spending decision on the client's behalf without the client: mitigation spends money to protect service, and the choice between expediting, drawing on stock, and accepting the delay belongs to the party paying for it, taken from the options presented.

The takeaway: bad news travels once, early, with a revised promise and options attached.

Practice Question 3

Stripping an import container, a warehouse team counts 96 cartons against 100 on the packing list. The supervisor proposes to write off four cartons immediately so the stock record can be corrected today. The records later show the seal intact and matching at every handover, and the strip sheet was marked only at the doors.

Which statement best describes the correct handling of this discrepancy?

1. The supervisor is right: the physical count is the truth, so the record should be adjusted to 96 at once and the four cartons claimed from the carrier.

2. The shortage must have occurred in transit because the goods were counted on arrival, so the carrier should be notified and the claim pursued before any internal recount.

3. No action is needed in either direction: the four cartons will surface at the annual stock count, and the variance can be absorbed into that count's tolerances.

4. The adjustment must wait: the discrepancy is traced through the records first and the physical recount completed, and only then is the file resolved — closing with root cause analysis and corrective and preventive action on the counting method, not merely a corrected number. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

The reconciliation reflex — records before adjustments — treats the discrepancy as a question. The chain reconstruction (a seal intact and matching at every recorded handover, in the manner of the security lesson) clears the transit by evidence, and the physical recount answers what the records leave open. Only when the count is verified does the record move, and the incident then closes in the response framework's final movement: root cause analysis (counting at a single point treated the doorway's view as the container's contents), corrective action (a tally per lift, a second count at clearance, the strip sheet signed before posting), and preventive action (the two-point tally extended to other receiving points). A corrected number alone closes nothing.

Answer 1 adjusts on an unverified count and launches a claim the evidence does not yet support — in the worked example, the four cartons were standing behind the last pallet row. Answer 2 assumes a transit loss that the unbroken seal sequence argues against, choosing blame before evidence. Answer 3 abandons the reconciliation entirely and lets a procedure error harden into an annual mystery.

The takeaway: investigate before you adjust, and close the incident with a cause and a change, not with a number.

Fact-check appendix (audit notes — not for entry)

Definitions of shipment exception, shipment delay, cargo damage, short shipment, over shipment, documentation discrepancy, incident report, exception management, root cause analysis, corrective action, preventive action, cargo survey, claim notification, and contingency plan — source: curriculum lesson brief module-11/lesson-11.4.md (key terms, taught as defined)

The risk-management cycle runs identify, assess (likelihood and impact), mitigate/treat, monitor and review, with communication throughout — source: research sheet logistics-scm-frameworks.md section 11

COVID-19 port congestion ran from early 2020 through roughly 2021–2022 (demand shifts, labour disruptions, container imbalances), kept qualitative — source: research sheet logistics-scm-frameworks.md section 11

The Ever Given ran aground in the Suez Canal on 23 March 2021 and was freed on 29 March 2021, blocking the canal for six days and seven hours with hundreds of vessels queued (at least 369 waiting by 28 March, over 400 by reopening) — source: research sheet logistics-scm-frameworks.md section 11 (Wikipedia, "2021 Suez Canal obstruction")

Attacks on shipping in the Red Sea from late 2023 led major container lines, from December 2023, to reroute widely around the Cape of Good Hope, substantially lengthening Asia–Europe voyages — source: research sheet logistics-scm-frameworks.md section 11 (Wikipedia, "Red Sea crisis"); no voyage-length percentages or loss figures used anywhere in the lesson

OTIF (on-time in-full) counts only deliveries both on time and complete; late or short fails — source: research sheet logistics-scm-frameworks.md section 7 (MetricHQ)

Safety stock buffers against supply and lead-time variability; the service level is the target probability of no stockout — source: research sheet logistics-scm-frameworks.md section 5 (taught generically, no formulas or benchmarks)

Stock reconciliation investigates discrepancies (mis-picks, unrecorded moves, theft, unit-of-measure errors) before adjusting records — source: research sheet logistics-scm-frameworks.md section 5

Total-cost trade-off: longer, slower transit raises in-transit inventory and carrying cost; inventory carrying cost is a real cost line even without an invoice — source: research sheet logistics-scm-frameworks.md sections 5–6; consistent with the Lesson 10.1 draft's carrying-cost treatment

Three-way match: purchase order, goods receipt, and supplier invoice agree before payment; discrepancies investigated first — source: research sheet logistics-scm-frameworks.md section 3

Clean transport document shows no remarks on apparent condition; a claused document notes defects — source: research sheet trade-transport-standards.md section 4 (applied generically to delivery receipts; no bank/letter-of-credit detail re-taught)

Insurers commonly require a survey of damaged cargo for substantial claims, and the survey report records condition, cause, and extent — source: consistent with research sheet trade-transport-standards.md section 10 context and the Lesson 11.1 draft's claim sequence; general insurance practice

An insurer having compensated may take over the claim against the party responsible (subrogation) — source: general business knowledge

Carrier liability is bounded by the carriage contract and modal liability rules, which attach their own notice time requirements — source: general business knowledge, consistent with the Lesson 11.1 draft; no liability figures, amounts, or legal deadlines stated anywhere in the lesson

Early detection from tracking and status events; revised ETA communication with cause and options — source: lesson brief module-05/lesson-5.4.md and its draft treatment

Receiving inspection before acceptance and dispatch verification — source: lesson brief module-08/lesson-8.2.md; safety stock and service level — lesson brief module-08/lesson-8.3.md; records-before-adjustments reconciliation — lesson brief module-08/lesson-8.4.md

The equipment interchange receipt records container and seal condition at interchange — source: lesson brief module-09/lesson-9.3.md (EIR key term)

Document set cross-checking before action — source: lesson brief module-06/lesson-6.4.md; forwarder's representative role — lesson brief module-05/lesson-5.1.md

All worked-example quantities, counts, timings, and outcomes are illustrative for teaching purposes per writing spec rule 5.1; the lesson contains no claim statistics, no carrier liability amounts, no insurance figures, and no statutory or contractual notice periods

Cross-references: Lesson 5.1 Role and Responsibilities of Freight Forwarders; Lesson 5.4 Shipment Tracking and Customer Communication; Lesson 6.4 Document Verification and Common Discrepancies; Lesson 8.2 Receiving, Put-Away, Picking, and Dispatch; Lesson 8.3 Inventory Control, FIFO, and FEFO; Lesson 8.4 Warehouse Documentation and Stock Reconciliation; Lesson 9.3 Cargo Release, Delivery Orders, and Terminal Procedures; Lesson 10.1 Components of Total Logistics Cost; Lesson 11.1 Cargo Insurance and Risk Allocation; Lesson 11.2 Dangerous Goods and Special Cargo Handling; Lesson 11.3 Supply Chain Security and Cargo Integrity — source: curriculum lesson briefs

Module 11 Assessment — 12 questions

Enter via the instructor portal's Configure assessment step. Green ticks mark correct answers — do not enter them; enter options as-is.

Question 1 (tests lesson 11.1)

1

A factory in Kumasi buys a shrink-wrapping line from an overseas machinery manufacturer on EXW terms. The factory engages a regional trucking firm, whose truck and crew arrive at the manufacturer's works to collect. The machine is placed at the buyer's disposal in the yard, and while the buyer's own crew is rigging it onto the buyer's truck, the machine slips from its lifting sling and is damaged. The factory demands that the manufacturer compensate the loss, arguing that the damage happened on the seller's own premises before the journey even began.

Which statement best describes the position?

1. The manufacturer must compensate, because the machine was still standing on the seller's premises when it was damaged, and responsibility stays with the seller until the goods physically leave the seller's ground.

2. The trucking firm must compensate the factory in full, because a carrier that has taken goods into its custody is automatically liable for the full commercial value of anything damaged in its hands.

3. Under EXW the risk passed to the buyer when the goods were placed at the buyer's disposal at the seller's premises, so the collection and the loading were at the buyer's risk; EXW imposes no insurance duty on the seller, and covering the collection leg was the buyer's own arrangement to make. ✓

4. The manufacturer must compensate, because a seller always insures the goods until collection is complete, so the seller's insurers must answer for damage occurring before the truck departs.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The lesson's three-step sequence runs: identify the risk, allocate it by the trade term, then ask who insures. Under EXW the risk passes to the buyer when the goods are placed at the buyer's disposal at the seller's premises. From that moment the machine's exposure — the collection, the loading, and the journey — belongs to the buyer, whose own crew and contractor were performing the lift. The party holding the risk on each leg is the party whose money is exposed and the natural buyer of cover for it, so arranging insurance for the collection was the buyer's own decision to take or skip; EXW, like every rule except CIF and CIP, contains no duty on the seller to insure at all.

Answer 1 moves the risk onto the physical ground: responsibility under a trade term follows the agreed transfer point, not the location where the loss happened, and the machine stood at the buyer's disposal before the lift began. Answer 2 overstates carrier liability, which is neither automatic nor unlimited — the carriage rules and the contract attach defences and limits — and the trucking firm here is the buyer's own contractor, engaged on the buyer's terms. Answer 4 invents an insurance obligation running until collection is complete; only CIF and CIP oblige a seller to insure, and neither the seller's premises nor the incomplete loading creates one.

The takeaway: fix the risk-transfer point before asking who pays and who insures; on an EXW collection the buyer owns the risk from the moment the goods stand at its disposal.

Question 2 (tests lesson 11.1)

2

An importing company in Tema buys two consignments from two overseas suppliers. The first, finished ceramic sanitary ware, is sold on CIP Tema terms. The second, a bulk consignment of a robust raw commodity, is sold on CIF Tema terms. Both sellers arrange insurance and send certificates of insurance, and on checking the two documents the company's clerk finds that both policies were written at Institute Cargo Clauses (C). The clerk asks whether the two covers comply with the sales' terms.

Which statement best describes the position of the two policies?

1. Both policies comply, because the seller under either term may select whichever ladder level best matches the cargo's character, since the Incoterms 2020 insurance duties name no particular level.

2. The CIF policy complies and the CIP policy falls short: under CIF the seller must insure at minimum Institute Cargo Clauses (C) for the buyer's benefit, while under CIP the required minimum is the higher (A) level, whatever the cargo's character. ✓

3. Both policies fall short, because CIF and CIP both impose Institute Cargo Clauses (A) as an absolute maximum that no policy may exceed without breaching the trade term.

4. The CIP policy complies and the CIF policy falls short, because sea-only CIF demands the widest cover while multimodal CIP, whose risks are spread across more carriers, demands only (C).

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

Only two of the eleven Incoterms 2020 rules contain a duty to insure: CIF and CIP. The duty fixes a minimum level for each. Under CIF the seller must insure at minimum Institute Cargo Clauses (C) for the buyer's benefit; under CIP the seller must insure at the higher (A) level, reflecting CIP's use across all modes and commonly for higher-value, damage-prone consignments. The bulk consignment sold on CIF is therefore correctly covered at (C), and the sanitary ware sold on CIP is not: whatever one thinks of the cargo's character, (C) sits below the minimum the CIP rule itself imposes. The duty is a floor and not a ceiling — the parties may agree more, and the cargo's character may justify more — but the term's minimum binds where the contract is silent.

Answer 1 dissolves the minimums by pretending the rules name no level; both rules name theirs, and the ladder exists precisely so the level can be stated. Answer 3 reverses the logic: the stated levels are minimums the parties may exceed, never maximums they may not. Answer 4 swaps the two duties, reasoning from mode breadth to cover breadth; the allocation is fixed by the rules themselves, and no sea-versus-multimodal argument lowers the CIP requirement.

The takeaway: under CIF check the certificate for (C) or better, under CIP for (A) or better — and the buyer's proof is the certificate of insurance the term entitles it to receive.

Question 3 (tests lesson 11.1)

3

A general cargo vessel touches bottom on a shoal while leaving a river port fully laden. To refloat her before the tide turns, the master orders part of the deck cargo jettisoned into the river. The vessel comes free, and both she and the remainder of the cargo complete the voyage safely. Aboard are consignments for two importers: Importer M, whose containers were lifted overboard in the jettison, and Importer N, whose containers below deck were never touched. On arrival, the vessel's owners declare general average, and Importer N is asked to post security for a contribution to the sacrifice.

Which statement best describes the outcome for the two importers?

1. Importer M bears its loss alone, because cargo deliberately jettisoned sits outside every sharing rule, while Importer N, whose goods were untouched, owes nothing to anyone.

2. The jettison was the master's unilateral act, so the vessel's owners alone must compensate Importer M in full, and the declaration cannot bind Importer N without its prior written consent.

3. The declaration is invalid because the voyage ended safely; general average can arise only where the vessel itself is lost.

4. Importer M claims its sacrifice through the general average adjustment, and Importer N contributes in proportion to the value its cargo brought safe to the adventure, because the sacrifice preserved the common adventure; cargo owners ordinarily post security for their contributions, typically through their insurers, before delivery. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

General average is the maritime principle, codified in the York-Antwerp Rules, under which a voluntary sacrifice, such as cargo jettisoned to lighten a vessel, or an extraordinary expense, deliberately made to preserve the common adventure from a common peril, is shared proportionally among the ship, the cargo, and the freight interests that the rescue saved. The jettison freed a vessel whose hull, remaining cargo, and earnings the sacrifice preserved, so every cargo interest contributes roughly in proportion to the value its cargo brought safe to the adventure. Importer M's claim for the containers lost overboard and Importer N's contribution are settled inside the same adjustment, and security is ordinarily required before the goods move. The lesson adds the ladder connection: general average sacrifice is a recognised peril covered even at the narrowest level, Institute Cargo Clauses (C).

Answer 1 applies ordinary own-loss intuition on both sides at once; the point of the sharing rule is that the rescue was made for everyone who benefited, the touched and untouched alike. Answer 2 demands a consent the principle never asks for; the master's authority to sacrifice in the common peril is exactly what general average exists to price. Answer 3 confuses general average with total loss; the rule operates when the adventure is saved, not when it is lost.

The takeaway: a rescue for one is a charge on all, and an untouched cargo owner's policy, even at (C), turns the demand into a filed problem instead of a funded one.

Question 4 (tests lesson 11.2)

4

An export clerk at a trading company near Tema Port prepares the company's first sea shipment of a newly reformulated industrial coating, packed in new UN-performance drums. Under time pressure, the clerk never obtains the product's safety data sheet. Working from the supplier's marketing flyer, which describes the coating as low-odour and user-friendly, the clerk marks each drum with the product's trade name, applies a flammable-liquid hazard label copied from the previous formulation's consignment, and completes a declaration giving the trade name only — no UN number, proper shipping name, hazard class, or packing group.

Which statement best identifies the first point at which the compliance chain failed, and the consequence?

1. The first failure is classification: the shipper's duty is to read the safety data sheet and classify the new formulation before anything else, because the packaging standard, the marks, the labels and the declaration are all built on that classification, and a declaration naming a trade name instead of a UN number and proper shipping name presents the chain's outputs without their foundation. ✓

2. The first failure is documentary: the consignment would be compliant if the supplier's marketing flyer were attached to the file, since the flyer can stand in for the missing safety data sheet.

3. The first failure is physical: new UN-performance drums make the consignment compliant whatever the papers say, because containment is what actually protects the handlers.

4. The first failure is at the terminal: responsibility passed to the carrier's trained acceptance staff the moment the container left the gate, so the shipper's file cannot affect the consignment's compliance afterwards.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

Classification is the foundation of dangerous-goods compliance, and the shipper owns it personally: the safety data sheet is read, the rulebook applied, and the results — the hazard class, the UN number, the proper shipping name, and the packing group where one applies — stand behind everything that follows. The performance standard of the packaging is selected for the classified hazard; the marks state the UN number and proper shipping name the classification produced; the hazard label announces the class; and the declaration declares those facts in the form the modal rulebook prescribes. A consignment classified from a marketing flyer has skipped the only step that generates the facts, so the marks, the label and the declaration present outputs without a foundation — and the declaration, the shipper's own statement, names no UN number or proper shipping name at all.

Answer 2 repeats the substitution error the lesson's closing note dismantles: the safety data sheet is the starting point for deciding whether goods are regulated, never a substitute for the declaration, and a marketing flyer is not a safety data sheet at all. Answer 3 separates containment from communication; packaging alone makes no consignment compliant, because the chain exists so that every handler and responder knows what is inside. Answer 4 misplaces the duty: the trained acceptance check is a gate that catches failures, and it presupposes — it does not replace — the shipper's work before the gate.

The takeaway: SDS first, classification second, declaration third, and nothing shipped until all three agree.

Question 5 (tests lesson 11.2)

5

A chemical distributor in Ghana imports two class 3 flammable liquids from the same overseas supplier. On the two dangerous goods declarations, solvent X is assigned packing group I and solvent Y is assigned packing group III. Reviewing the file, a new member of staff asks what the difference between the two entries means for the operation.

Which statement best describes what the packing groups record and change?

1. The entries record package size: solvent X travels in larger drums and solvent Y in smaller ones, and the distributor should order its packaging to match the group number.

2. The entries are the supplier's commercial gradings with no operational meaning; only the hazard class, class 3 in both cases, affects packaging and stowage decisions.

3. Solvent X presents greater danger within its class than solvent Y: the packing group grades the degree of danger the substance presents — I denoting great danger, II medium danger and III minor danger — and drives the performance standard the packaging must meet, together with the stowage and segregation care around the consignment. ✓

4. Solvent Y presents the greater danger, because the higher numeral denotes the higher danger, in the same way that the hazard class numbers climb from 1 to 9.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

A packing group is a statement about the substance: within its class, packing group I denotes great danger, II medium danger, and III minor danger. Solvent X is therefore graded the more dangerous of the two class 3 liquids, and the grading does real work downstream: it selects the performance standard the UN-performance packaging must meet, it appears on the dangerous

goods declaration, and it informs the stowage and segregation decisions taken around both consignments. It grades how dangerous, never how big.

Answer 1 is the misreading the lesson isolates early: the group determines the performance the packaging must deliver, not a drum size, which follows the product and the journey. Answer 2 strips the entries of meaning; the packing group is precisely the instrument that distinguishes two substances sharing one class, which is why it travels on the documents. Answer 4 inverts the scale: the ladder's letters climb with protection, but the packing-group numerals run the other way, I above II above III, and nothing in the rulebooks reads group III as more dangerous than group I.

The takeaway: read the packing group as the substance's danger rating within its class — I the greatest, III the least — and let it set the packaging standard and the care around the consignment; and remember that for classes 1, 2 and 7 no packing group exists at all.

Question 6 (tests lesson 11.2)

6

A freight forwarder is consolidating a forty-foot container at a container freight station for a regional sea service. Already booked inside is a documented consignment of a class 5.1 oxidising substance. A late booking arrives: drums of solvent-based printing ink documented as class 3, flammable liquids, which would exactly fill the container's last gap. The warehouse supervisor, with many years of experience at the station, proposes to load the ink: both consignments are sealed and sound, the pairing has been carried many times over the years without incident, and the space would otherwise sail empty.

Which statement best describes the position?

1. The supervisor's experience is the appropriate authority here: segregation decisions rest on personal professional judgment, and years without incidents are practical evidence that this pairing is safe.

2. The consolidation check is documentary and comes before packing: the classes present are checked for compatibility under the applicable requirements before the container is packed, because a flammable liquid and an oxidising substance each make the other's failure worse; many quiet voyages in the past are not evidence about the next one. ✓

3. Segregation requirements govern cargo on open vehicles and decks; inside a closed container any pairing may be stowed as space allows, because the container walls separate the goods.

4. The pairing would be acceptable if the two consignments travelled on separate pallets, since segregation requires only that incompatible goods do not touch.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

Segregation is the separation of incompatible dangerous goods according to applicable safety requirements, and the flammable-liquid-and-oxidiser pairing is the classic incompatibility the lesson teaches: one supplies the fuel for a fire, the other the oxidiser that feeds it, and sealed together in a container at sea each makes the other's failure worse. The check is procedural and documentary, and it has a place in the process: before the container is packed, while the plan is still on paper, the classes present are checked for compatibility under the applicable requirements — the same consolidation discipline that treats groupage as a documented checking process rather than a loading exercise. Caught at that moment, the fix costs a re-booking and a telephone call instead of a stowage decision nobody would defend afterwards.

Answer 1 substitutes experience for the check. Long experience of quiet voyages is exactly the trap the lesson's counterfactual dismantles: most voyages pass without incident, which is why incompatible consolidations accumulate — the absence of disaster reads as proof of safety until the day a leaking drum and heat in the hold meet an oxidiser. Answer 3 promotes the container walls into a segregation solution; the closed container is precisely the enclosed space in which the combination is worst. Answer 4 reduces segregation to touching, which the applicable requirements do not do.

The takeaway: the consolidation check is documentary, it happens before packing, and it is never satisfied by years of nothing having happened yet.

Question 7 (tests lesson 11.3)

7

At a delivery to a warehouse in Kumasi, a driver comparing the container's seal number against the delivery documents finds that the high-security seal on the doors does not match the number recorded on the bill of lading. A colleague recalls that the container was opened for a customs examination while in transit — and the journey file contains a dated record of that examination, showing the authority for the opening, the reason, the time and place, and the number of the replacement seal applied, all recorded at the time.

Which statement best describes how the driver should treat this position?

1. The mismatch is explained by a documented story: the record shows who broke the original seal, on whose authority, why, and which replacement seal was fitted, so the consignment may proceed to its normal checks and stripping, since a seal broken with a record is not tampering. ✓

2. Any difference between the seal on the doors and the bill of lading makes the container evidence in every case, and even a documented customs examination cannot legitimise it, so the delivery must be refused outright.

3. The colleague's recollection is sufficient: a verbal account of a customs stop witnessed by the crew settles the matter, and the container may be opened and stripped on that understanding.

4. The driver should cut off the replacement seal, fit a new seal of his own, note the new number on the delivery documents, and complete the delivery so the vehicle is not delayed further.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

The seal discipline treats a seal that disagrees with the documents as a stop event — but the lesson's second boundary matters here: the discipline does not forbid breaking a seal, it forbids breaking one without a story. Whenever a seal must be broken and replaced before delivery, the event is documented at the time: who broke it, on whose authority, why, when and where, who witnessed it, and the numbers of both the old seal and the new. The transit examination's record carries exactly those elements, so the mismatch on the doors is explained continuity rather than an unexplained story, and the delivery proceeds to its normal checks.

Answer 2 overreaches in the opposite direction: a documented, authoritative break is not tampering, and treating every recorded examination as a refusal would break legitimate trade. Answer 3 accepts memory in place of record; a recollection is a story told after the fact, and the lesson's rule is that records written at the time locate the break in the chain while a story told later locates nothing — to the next stranger who checks the seal, an undocumented replacement is indistinguishable from tampering. Answer 4 performs the silent break-and-replace the discipline condemns: it destroys physical evidence and fabricates documentary continuity.

The takeaway: a seal may be broken, but never silently; the story must be written down at the time, with both numbers, so that continuity — not contradiction — faces the next checker.

Question 8 (tests lesson 11.3)

8

Two exporters ship consignments on the same vessel. After discharge overseas, prohibited items are found inside both containers, inserted after packing by other parties in the chain. Exporter P supervised its packing, verified the packed goods against the packing list before sealing, applied

and recorded a high-security seal at packing, and handed the container onward through signed handovers to vetted carriers. Exporter Q's container was packed by an unsupervised casual crew, and Q cannot show what was checked, when the container was sealed, or who handled it afterwards.

Which statement best describes the two exporters' positions?

1. The two positions are identical, because a shipper answers for everything found in its container whatever it did, and nothing it can show makes any difference.

2. Exporter P is exempt from all scrutiny, because a recorded seal is a guarantee that the container was never opened, while Exporter Q is treated as the importer of the prohibited items.

3. Neither exporter can be questioned, because responsibility for goods inserted after packing lies with parties outside the trade chain altogether.

4. Exporter P can use its records — supervised packing, the pack verified against the list before sealing, the seal applied and recorded at packing, and signed handovers to known parties — to show what it packed and where its control ended, so an insertion can be demonstrated to have happened to it; Exporter Q, unable to account for its own load, cannot draw that line. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

The container carries the shipper's name and the shipper's declaration, and if prohibited goods inserted after packing are found inside, the discovery happens against that shipper first — to customs, to the carrier, and to the buyer. Innocence is therefore an outcome the shipper must be able to demonstrate, not a status anyone grants on request. Exporter P's controls are the demonstration: supervised packing means the shipper knew what entered the container; the pack verified against the packing list fixed the record at the moment of greatest knowledge; the high-security seal applied and recorded at packing froze that moment into evidence; and signed handovers to vetted parties protected the evidence along the chain. Those records let P show what it packed and where its control ended, so the insertion can be demonstrated to have happened to it rather than because of it.

Answer 1 is the fatalism the lesson exists to prevent: identical discoveries are not identical positions, because one shipper can produce the trail and the other cannot. Answer 2 overstates the seal into an absolute guarantee; any seal can be broken, and its number is evidence, not a certificate of inviolability. Answer 3 invents strangers to the transaction; the parties who must be answered are the ones whose systems carry the consignment.

The takeaway: integrity discipline is self-protection — it is how the honest operator proves it was honest when someone else was not.

Question 9 (tests lesson 11.3)

9

Late on a Friday, the operations desk of a logistics company in Accra receives an email appearing to come from a long-standing overseas customer, instructing that a container due for delivery next week be redirected to a different warehouse, with new collection contact details. The email is written in the customer's usual style and signs off with a name the desk recognises.

Which response best reflects the discipline the lesson teaches for instructions and data?

1. Act on the instruction without delay: the writing style and a familiar signature are sufficient identification, and holding up a confirmed customer's instruction would damage the service relationship.

2. Verify the instruction by calling the customer on a number taken from the company's own records before anything is acted on, and let the goods move only when the instruction, the record, and the physical reality agree; an emailed change of delivery details is verified, never assumed. ✓

3. Reply to the email asking the sender to confirm the change, and act once a reply arrives, since the email exchange itself establishes who sent the instruction.

4. Decline every change of instruction once a consignment is moving, since delivery details can never be altered after dispatch under any circumstances.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

Documentation and data fraud is mostly impersonation: falsified release or delivery instructions, manipulated consignment details, documents altered after issuance. The controls are procedural. Instruction verification requires that a change of delivery details received by email be verified by calling the counterparty on a known number — one taken from the company's own records, not from the suspect message — before it is acted on. The three-way match states the same logic generally: instruction, physical reality, and record must agree before goods move. Call-back verification costs minutes; a diverted load costs the load.

Answer 1 authenticates by style and a familiar signature, which is exactly what a competent impersonation copies; nothing inside the email itself proves who sent it. Answer 3 verifies the instruction through the instruction's own channel: a reply from the same address, or the same

compromise, confirms nothing, and an email exchange is not the independent second source the discipline requires. Answer 4 overcorrects into a rule no operation can run on; changes of instruction are legitimate and often urgent, and the discipline governs how they are verified, not whether they may exist.

The takeaway: an emailed change of delivery details is verified on a known number before a wheel turns — and instruction, record, and reality must agree before goods move.

Question 10 (tests lesson 11.4)

10

At a warehouse yard near Tema, a delivery of assorted industrial goods is being unloaded when a drum on one pallet is found leaking a flammable liquid, and several cartons beneath it are soaked. The yard supervisor must direct the team's first actions.

Which sequence best reflects the incident-response framework the module teaches?

1. Notify the insurer and the carrier first and hold the scene exactly as it is until their instructions arrive, since commercial notification must precede every operational step.

2. Photograph the drum, the pallet positions and the soaked cartons in detail first, then stabilise the leak, since evidence must be preserved before anything else is disturbed.

3. Make safe first — protect people and stabilise the leaking drum — then preserve the evidence as found: photographs before anything is moved or cleaned, the damaged goods and packaging retained and segregated, nothing repaired or repacked; then notify the parties each within its own process — the forwarder and carrier, the insurer by claim notification, and the authorities a dangerous-goods incident brings in, with safety-related notification preceding commercial steps. ✓

4. Repack the undamaged stock into sound cartons to protect its saleability, clean the yard before the spill spreads, and file the incident report once the outcome of any claim is known.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The incident-response framework has six movements in order: detect early, contain and make safe, preserve evidence, notify the right parties, resolve and record, and learn and improve. The leaking flammable drum invokes the second movement's absolute priority — safety precedes commerce, always — and the dangerous-goods handling taught in Lesson 11.2 governs the stabilisation before any claim is discussed. Only then is evidence preserved: photographs before

anything is moved or cleaned, the damaged goods and their packaging retained and segregated, nothing repaired or repacked — because the surveyor surveys what is preserved, and a scene tidied is evidence destroyed. Notification follows, each party within its own process: the forwarder and carrier, the insurer by claim notification while the evidence is fresh, and the authorities a dangerous-goods incident brings in — with safety-related notification preceding commercial notification.

Answer 1 holds a live flammable leak hostage to claims instructions and inverts the ordering: commercial notification is one movement among several, and none precedes containment. Answer 2 photographs first, treating evidence as more urgent than the people in the yard; a stabilised scene can still be photographed, and a protected scene is itself the stronger evidence. Answer 4 performs the exact destruction the lesson's Important Note condemns — repacking, cleaning, and withholding the report until the claim's outcome — converting an investigable incident into an unexplained loss.

The takeaway: make safe, preserve as found, then notify every channel it concerns — the order is the discipline.

Question 11 (tests lesson 11.4)

11

A warehouse team in Tamale strips an import container of packaged sanitary fittings and counts 104 cartons against 100 on the packing list and the commercial invoice. A colleague proposes to book the four extra cartons into stock at once as a windfall, so that the record balances today, arguing that no one has lost anything and the matter needs no further work.

Which statement best describes the correct handling of this finding?

1. The surplus is a discrepancy like any other: it is traced through the records — the invoice, the packing list, the purchase order, the supplier's dispatch papers — and regularised before any stock figure moves, because undeclared extra goods create record and customs discrepancies that must be declared and corrected, not quietly absorbed. ✓

2. The colleague is right: an over shipment is a gain with no injured party, so increasing the stock record by four closes the matter, and discrepancies matter only where goods are missing.

3. The container should be resealed at once and returned to the carrier unopened, since any surplus is necessarily a carrier's mis-delivery and must go back before it is counted.

4. The surplus should be left to surface in the annual stock count's variances at year end, since customs requirements apply only to shortages, never to extra goods.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

A discrepancy is any disagreement between what the records say and what the goods show, and the lesson is explicit that the definition runs both ways: an over shipment is not a windfall but a shipment containing more goods than the records specify, creating record and customs discrepancies that must be declared and regularised, not quietly absorbed. The reconciliation reflex — records before adjustments — treats the surplus as a question the documents must answer: checked across the invoice, the packing list, the purchase order, and the supplier's dispatch papers, and traced before any stock figure moves. Only on the strength of that investigation does the record change — perhaps an invoice correction follows, perhaps cartons belonging to another order are found mixed into the load — and the incident then closes, as every incident should, with a root cause and corrective and preventive action on the process that let the count disagree.

Answer 2 books the surplus as gain: it treats discrepancies as losses only, which the over-shipment teaching refutes, and it hides an unexplained variance inside a stock figure. Answer 3 destroys the investigation before it begins, returning the goods uncounted and unexplained. Answer 4 postpones the question into the annual count's tolerances and rests on a falsehood — customs requirements do not apply only to shortages.

The takeaway: investigate before you adjust, in either direction; a record that balances today by absorbing a question has stored the question, not answered it.

Question 12 (tests lesson 11.4)

12

In March 2021, the Ever Given ran aground in the Suez Canal and blocked the waterway for six days and seven hours while hundreds of vessels queued. A freight forwarder in Accra has several clients' consignments aboard vessels held in that queue, and the delivery dates promised to those clients are now unachievable.

Which response best reflects the way the module teaches a systemic delay to be managed?

1. No communication is owed: nobody in the office caused the grounding, and clients watching the news will understand the schedule consequences without a message.

2. Each affected client should be told that the forwarder's own performance is unaffected and that any delay is a matter between the client and the vessel's owners, whom the client should pursue directly.

3. Every affected booking should be moved to air freight immediately, with the revised invoices sent before any notification, so that the promised dates are protected whatever the cost.

4. The affected bookings are identified, and each client is re-planned early — the named, dated cause, a revised estimated arrival built on the expected reopening, and the options: hold for the new date, supply from alternative stock, or expedite a critical portion by a faster mode — while mitigation is chosen with total-cost eyes and the event is recorded, so that quotations, transit-time assumptions and the contingency plan are updated rather than silently returning to the old map. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

The lesson names the Ever Given grounding — 23 March 2021, freed on 29 March 2021, six days and seven hours of blocked waterway, hundreds of vessels queued — as proof that delay is systemic rather than personal failure: no trader caused it, and the delays arrived anyway. The professional response is a method, not an apology. Detect: pull the affected bookings against the queued sailings. Re-plan the promise early: each client receives the same three-part message — the named, dated cause, a revised estimated arrival built on the expected reopening, and the options — hold for the new date, supply from alternative stock or safety stock, or expedite a critical portion by a faster mode. Mitigate with total-cost eyes: extra transit keeps capital afloat longer, so choices such as air-freighting a critical portion are made by the party that pays, from the options presented. Record: the event, the decisions, and the permanent update to quotations, transit-time assumptions, and the contingency plan, so the operation does not silently return to the old map.

Answer 1 converts a systemic event into silence; clients building production and sales plans around the promise must not discover the delay from their own dock. Answer 2 deflects instead of managing, leaving clients without revised dates or options. Answer 3 spends a client's money unilaterally and notifies afterwards; mitigation is offered, not imposed.

The takeaway: when the system shifts, re-plan the promise before it is missed — and never let the assumptions behind it quietly return to the old map.